

White Label Farm App

Build Specification

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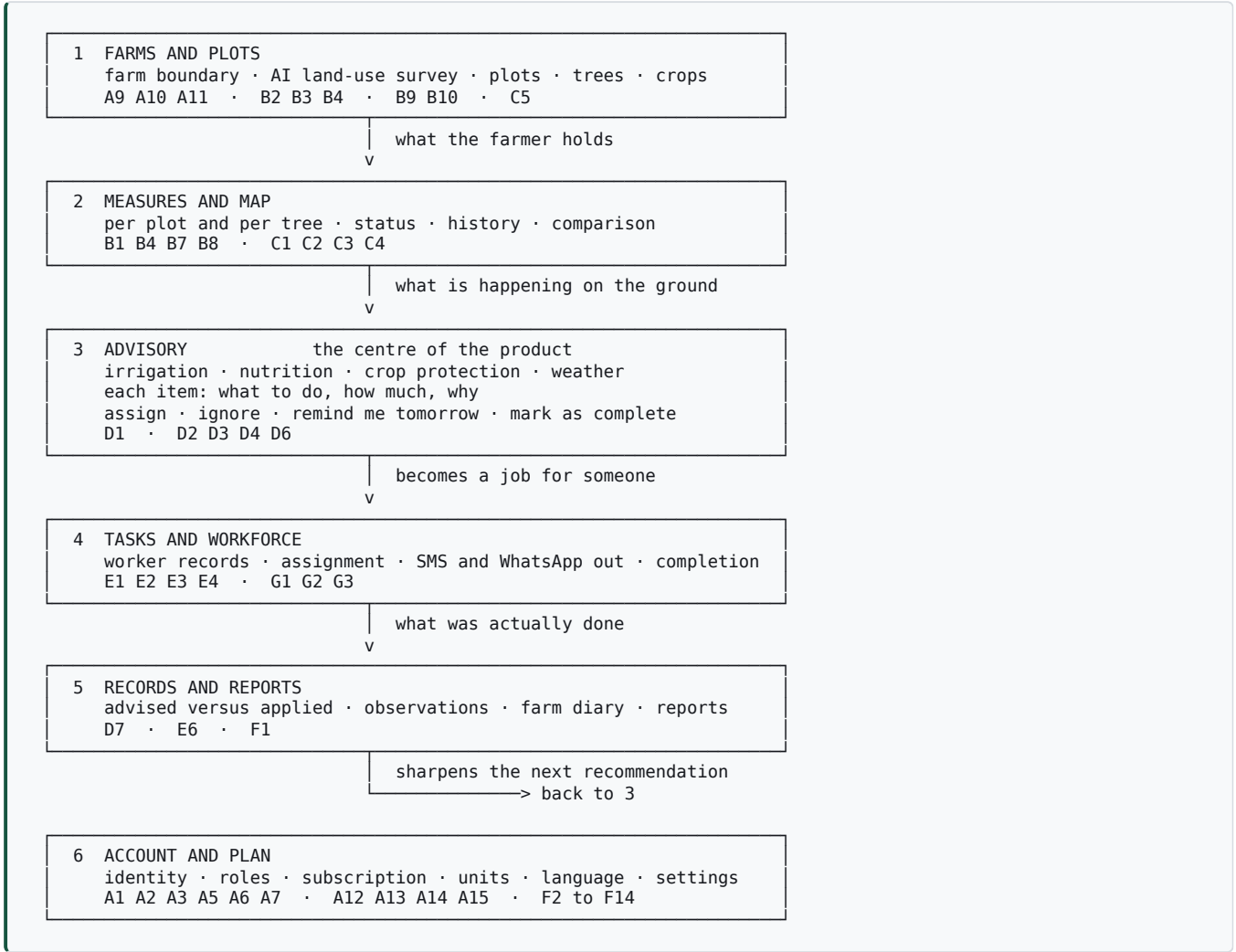
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1 Summary

1.1 Schematic of the app



1.2 At a glance

Product	White label farm management application, built on Map My Crop's satellite analytics platform
Markets	GCC and Jordan
Platforms	iOS 15+, Android 8.0+. Mobile only
Languages	Arabic, English, Hindi, Bengali, Pashto.
Roles	Farm Owner, Farm Supervisor, Farm Worker. Using a capability matrix
Workforce	Workers do not need the app. The owner holds worker records and reaches them by SMS or WhatsApp
Plans	Two levels per family — basic and pro — for crops, for trees, and combined. Priced from what the AI survey finds

Payment	In-app purchase for users registering in the app; web purchase with a server-side entitlement for users signing in by email; volume purchase for government and bulk
Hosting	Client data resides in the country it originates from
Offline	Read cached data anywhere; capture tasks, photos, observations and applied inputs offline
Farms	Up to 10 per account, on one subscription

2 General rules

2.1 Requirement identifiers

Every requirement carries an ID in the form **WFc.nnn**, where *c* is the chapter it belongs to and *nnn* its position within that chapter. Use these IDs in estimates, in commit messages, in test cases and in acceptance sign-off. The full register is in the last annex.

Chapter	Requirements
2 General rules	WF2.001 – WF2.021
3 Information architecture and navigation	WF3.001 – WF3.005
4 Onboarding and access	WF4.001 – WF4.117
5 Screen specifications	WF5.001 – WF5.194
6 Advisory layer	WF6.001 – WF6.028
7 Notifications	WF7.001 – WF7.009
8 Roles and permissions	WF8.001 – WF8.012
9 Plans and entitlements	WF9.001 – WF9.036
10 Internationalisation (i18n)	WF10.001 – WF10.022
11 Offline behaviour	WF11.001 – WF11.013

2.2 Terminology

Term	Meaning
Farm	The top-level land holding. Has a name, a location and a type.
Farm boundary	The outer polygon of a farm. Everything inside it belongs to that farm.
Plot	A mapped area of cultivated ground inside a farm boundary. Carries its own boundary. Analytics are computed per plot.
Trees	Tree stock inside a farm boundary. Analytics are computed per tree as well as per plot.
Survey	The land-use analysis MMC runs on a farm boundary, returning plots and trees classified by type.
Crop cycle	A time-bounded planting of one crop in one plot, from sowing to harvest. A first-class record with its own history — not an attribute of the plot.
Measure	A satellite-derived vegetation or health value (NDVI, NDRE, EVI, MSAVI and others). Called <i>measure</i> everywhere in the interface; the technical name appears only in the legend and in reports.
Recommendation	A piece of advice produced by the platform and shown to the user.
Task	A unit of work assigned to a person, with a due date and a completion record.
Observation	A field-captured note, usually with a photo and a GPS position.

Term	Meaning
Entitlement	The server-held record of what an account is allowed to see and do.
Worker record	A person the owner can assign work to. Name, phone number, language and notification preference. Does not require the person to install the app.

2.3 Global rules

These apply to every screen.

WF2.001 — Mobile application for iOS 15+ and Android 8.0 (API 26)+.

WF2.002 — Usable on a 5-inch screen at 360 × 640 dp.

WF2.003 — Install size below 80 MB.

WF2.004 — Minimum touch target 48 × 48 dp, with at least 8 dp between adjacent targets.

WF2.005 — The primary action on every screen sits in the bottom third of the display. Destructive or rarely-used actions go to the top.

WF2.006 — Body text no smaller than 16 sp. Numbers the user acts on (water volume, dose, tree count) no smaller than 20 sp.

WF2.007 — Respect the operating-system font-size setting up to 200% without clipping text or losing buttons off-screen.

WF2.008 — Status is not to be communicated by colour alone. Every colour-coded state also carries a symbol and a text label.

WF2.009 — Health and status use the scale below, with the same colour, symbol and word every time. Colour and symbol always travel together.

State	Colour	Symbol	Label (EN)	Meaning
Good	dark green	filled circle	Good	No action needed
Watch	light green	half circle	Watch	Monitor, no action yet
Needs attention	amber / orange	triangle	Needs attention	Act within days
Urgent	red	filled triangle	Urgent	Act today
No data	grey	dashed circle	No data	Cloud cover, no imagery, or outside subscription

WF2.010 — No screen presents more than one primary action.

WF2.011 — Every screen that can potentially have nothing to show (e.g. no farms yet, no plots on a farm, no tasks today) has a designed empty state: an icon, one line of plain language saying why it is empty, and a button that resolves it where one exists ("Add your first farm").

WF2.012 — Every screen has an explicit loading state and an explicit error state with a plain-language message and a *Try again* button. Log the error code, show the sentence.

WF2.013 — Numbers are always shown with their unit, rounded to actionable precision: "412 m³" instead of "411.8734 m³".

WF2.014 — Icons carry a text label beneath or beside them. Exception for obvious map controls and 3 dots “more” action.

WF2.015 — Never block the user with a full-screen “no connection” message when cached data exists. Show the cached data with its age and let the user continue.

WF2.016 — Every data-entry action that can be performed offline shows immediate local confirmation.

WF2.017 — The application carries Wafra branding only : logo, colours, app icon, splash, store listing and generated reports. Colour is applied through named design tokens, not hard-coded values.

2.4 Where data lives

WF2.018 — **Client data resides in a secure cloud in the country it originated from.** Emirati farm data stays in the UAE; Saudi farm data stays in Saudi Arabia. This covers farm boundaries, imagery derivatives, advice, tasks, photographs and personal data.

WF2.019 — Jurisdiction is resolved from the farm, not from the user. A supervisor in one country working a farm in another does not move that farm's data.

WF2.020 — Backups and disaster-recovery replicas are held under the same residency terms as primary storage.

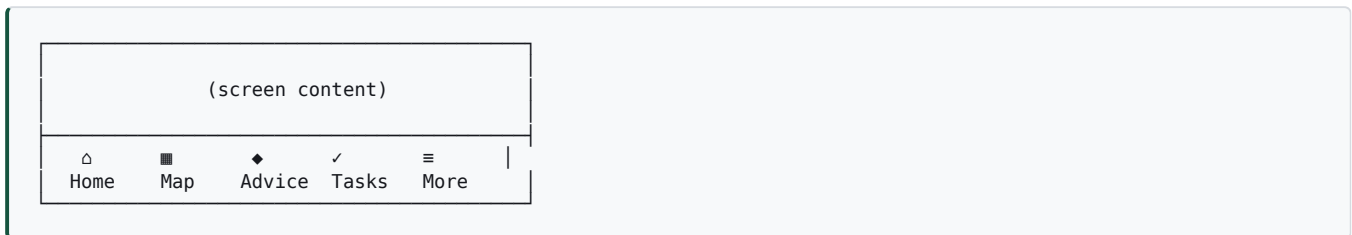
WF2.021 — Where a new market is opened, residency for that market is settled before the first farm is created in it, not afterwards.

3 Information architecture and navigation

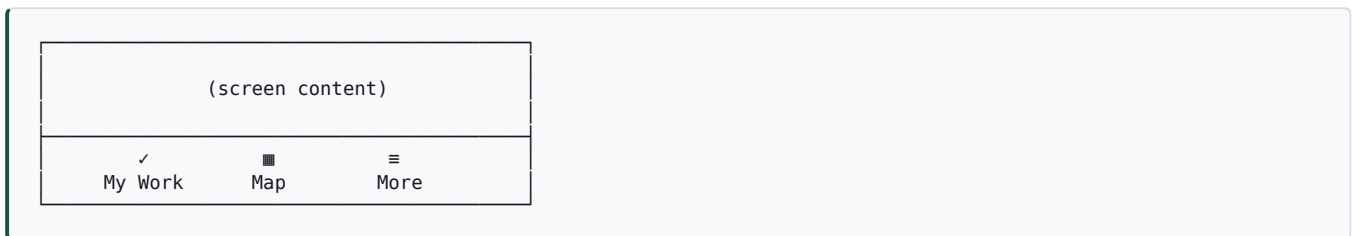
3.1 Global navigation

Navigation is a bottom tab bar, always visible, never hidden on scroll. Tabs are role-adaptive.

WF3.001 — Owner and Supervisor see five tabs:



WF3.002 — Worker sees three tabs, and lands on My Work at launch:



WF3.003 — A badge on the Advice tab shows the count of unread recommendations at **Needs attention** or **Urgent** severity only. Routine information is never badged.

WF3.004 — A badge on the Tasks / My Work tab shows the count of tasks assigned to this user and due today or overdue.

WF3.005 — Tab bar labels are always visible.

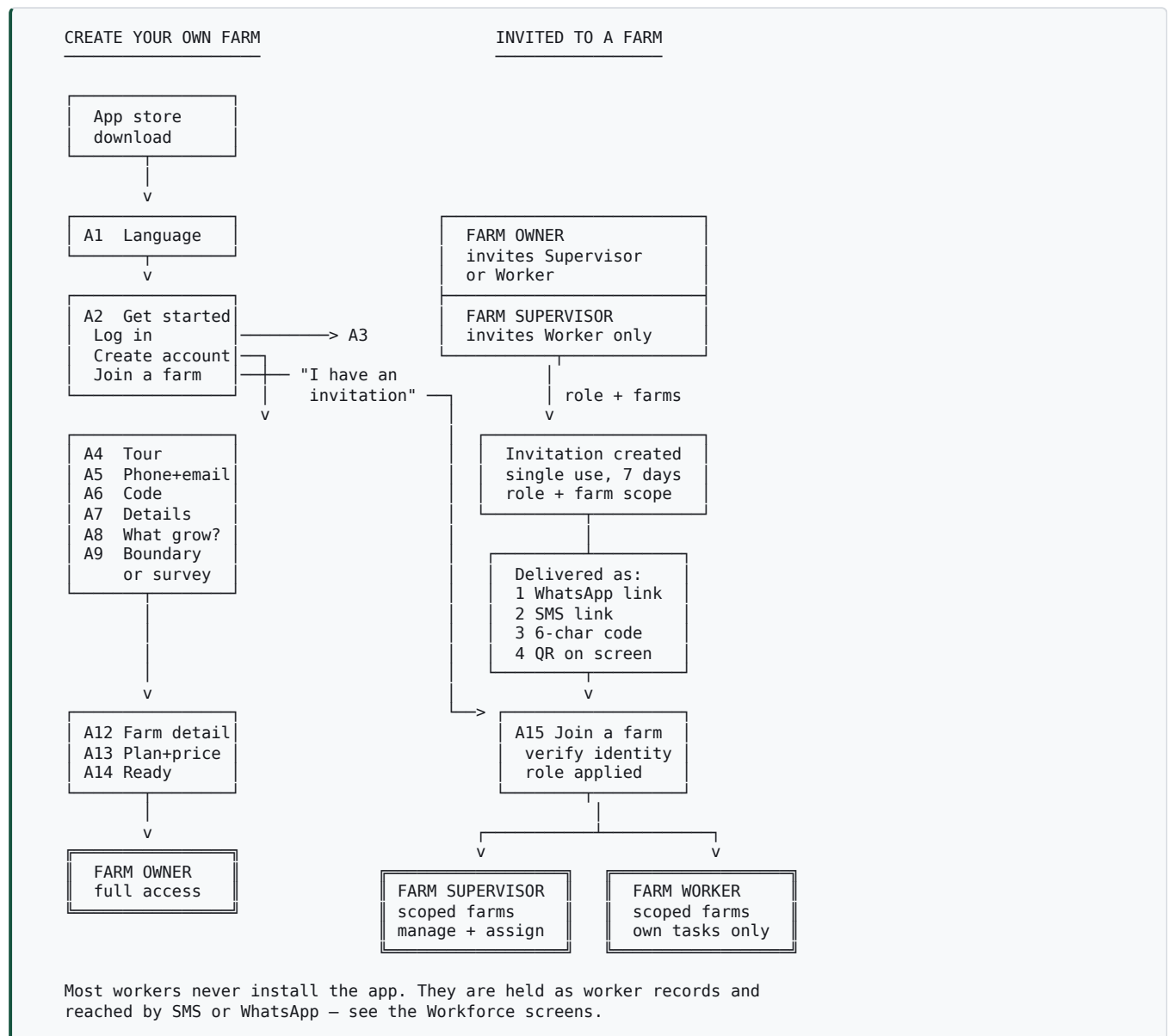
3.2 App Map

White Label Farm App		
First run		
A1	Language	
A2	Get started	log in · create account · join a farm
A3	Log in	identifier, code or password
A4	Guided tour	snapshots, new users only
A5	Sign up – phone and email	
A6	Verify code	
A7	Your details and units	
A8	What do you grow?	
A9	Add your first farm	
A9	Draw my plots myself	one or many
A10	Survey my whole farm	
A11	What we found	split, join, edit, remove, add
A12	Farm details	
A13	Your plan and price	
A14	You're ready	
A15	Join a farm (invitation)	
HOME (Owner, Supervisor)		
B1	My farms	all farms, or one farm
B2	Farm detail	
B3	Plots	
B4	Plot detail	
B5	Crop cycle detail	
B6	Add / edit crop cycle	
B7	Measure viewer (full screen)	
B8	History and comparison	
B9	Tree list	(tree and mixed farms only)
B10	Tree detail	
G1	Workforce	people who do not use the app
G2	Add a worker	
G3	Worker record	
B11	Farm settings	
B12	Add farm	
MAP (all roles)		
C1	Map	persistent search, locate a tree
C2	Layers	
C3	Plot sheet (bottom sheet on tap)	
C4	Compare dates	
C5	Boundary editor	
ADVICE (Owner, Supervisor)		
D1	Advice inbox	
D2	Irrigation advice detail	
D3	Nutrition advice detail	
D4	Crop protection advice detail	
D6	Weather alert detail	
D7	Record what you did	
TASKS (Owner, Supervisor) / MY WORK (Worker)		
E1	Task list	ADD button, bottom right
E2	Task detail	
E3	New task	
E4	Complete task	
E6	Field observation (scouting)	
E7	Photo disease check	
MORE (all roles, contents filtered by role)		
F1	Reports	
F2	Team and access	people who use the app
F3	Invite someone	
F4	Member detail	
F5	Subscription	
F6	Compare plans	
F7	Settings	
F8	Language, units and region	
F9	Notifications	
F10	Data and storage	
F11	Activity log	(Owner only)
F12	Help and user guide	
F13	Contact Wafra	
F14	My profile	

4 Onboarding and access

4.1 Registration and access model

Everyone registers with a mobile number **and** an email address, either of which can then be used to sign in. Which of the three paths a person takes is chosen at A2.



4.2 Roles

WF4.001 — Registration creates an account with **no role**. A role is attached to a user only through a farm: either by creating one, or by accepting an invitation to one.

WF4.002 — A user who creates a farm becomes a **Farm Owner** of it.

WF4.003 — A user who accepts an invitation receives the role carried in that invitation — **Farm Supervisor** or **Farm Worker**. The role is set by the inviter and is never chosen by the invitee.

WF4.004 — The same person may do both. Registering, creating a farm, and later accepting a Supervisor invitation to someone else's farm is a normal case: roles are held per farm, not per account.

WF4.005 — A Worker can never create, delete or edit a farm, a plot or a boundary, on any farm, however they arrived.

WF4.006 — An invitation is single use, expires after 7 days, and is bound to the role and the farm scope chosen at creation.

WF4.007 — Four delivery methods, all producing the same single-use token: WhatsApp deep link (primary), SMS deep link (fallback), a 6-character code the invitee types manually, and a QR code displayed on the inviter's screen.

WF4.008 — Inviting someone into the app is a different thing from creating a **worker record** for someone who will never install it. Both are available to an Owner; see §5.6.

WF4.009 — An Owner sees all pending invitations for their farms with expiry and a one-tap revoke. A Supervisor may invite Workers only, and only to farms they already have access to.

WF4.010 — Where a Worker has no smartphone of their own, a Supervisor can complete the invitation on the Worker's device by entering the 6-character code.

4.3 A1 — Language

Shown on very first launch. Nothing precedes it: the app cannot address anyone until it knows which language to do it in.



WF4.011 — Each language is shown **in its own script only**. No English name underneath and no second line of any kind.

WF4.012 — **Arabic is first** in the list. The remaining four follow.

WF4.013 — All five options fit on one screen without scrolling on a 360 × 640 dp display. Row height is tightened to whatever that requires.

WF4.014 — Pre-selected to the device's OS language if it is one of the five; otherwise Arabic.

WF4.015 — Selecting a right-to-left language mirrors the layout immediately, on this screen, before Continue is pressed.

WF4.016 — Never shown again after first run. Changed afterwards in Settings (F8) and on the login screen.

4.4 A2 — Get started

Three doors, nothing else. No fields, no keyboard, no small print. The app needs to know which of three people it is talking to before it asks anything.



WF4.017 — A2 carries **exactly three actions and no input of any kind**. No number field, no password field, no checkbox, no terms line.

WF4.018 — The three routes: *Log in* → A3, *Create an account* → A4, *Join a farm* → A15. Each collects its own inputs on its own screen.

WF4.019 — *Log in* is first, because the operating system usually remembers the credentials and the returning user is the commonest case after launch week.

WF4.020 — A language control stays on this screen, so a wrong choice at A1 costs one tap.

WF4.021 — Opening an invitation deep link goes straight to A15 and never shows A2.

4.5 A3 — Log in

← Log in

Mobile number or email

Send me a code

or

Password

Log in

Forgot your password?

WF4.022 — Mobile number and email are interchangeable identifiers. One field accepts either, and the app decides from the input whether to send the code by SMS or by email.

WF4.023 — Login accepts identifier + password, or identifier + one-time code. Both are offered on the same screen; the code is the more prominent option.

WF4.024 — Sessions persist for 180 days of activity on a personal device. Biometric unlock is offered where the device supports it.

WF4.025 — *Log out* is always available and clears the local cache including cached imagery and queued photos, after warning the user if anything is unsynced.

4.6 A4 — Guided tour

Shown after the user chooses *Create an account*, in the language they picked at A1. By this point the app knows they are new and knows how to talk to them — which is why the tour sits here and not before either.

It is a **sequence of snapshots**, not a live interface with banners over it, and not a demo account.

Skip

[satelkite snapshot of a typical farm, trees and crops outlined and named]

This is your farm from space

We find your trees and your fields and watch them every week.

1 of 5

Next

WF4.026 — The tour is a fixed sequence of still images with a caption. No live data, no account, no interactive screens dressed as a demo.

WF4.027 — The first snapshot is a **satellite view of a typical farm** with trees and crops identified on the image, so the value of satellite analysis is obvious before a word is read.

WF4.028 — The following snapshots are advisory examples, numbered *1 of 5*, *2 of 5* and so on, so the user knows how long the sequence is.

WF4.029 — *Skip* is present on every snapshot and goes straight to A5.

WF4.030 — The tour is shown once, on the registration path only. A returning user logging in never sees it. It is repeatable at any time from Help (F12).

WF4.031 — The tour runs in the language chosen at A1, from the same string catalogue as the rest of the app (§10.2).

4.7 A5–A6 — Sign up

WF4.032 — Registration takes **both a mobile number and an email address** for an account that will own a farm. Either can then be used to sign in.

← Create your account

Mobile number *

+966 5X XXX XXXX

Email address *

Both can be used to sign in

☐ I agree to the Terms of Use and Privacy Policy

Send code

WF4.033 — Email is **mandatory on the Owner role**. It is what makes the web purchase path possible (§9.1.3). A Worker record created by an owner needs a phone number only.

WF4.034 — Verification (A6) is a code, by SMS to the number or by email to the address, verified before the account proceeds.

WF4.035 — Country selector defaults from the SIM's country, then device locale, then Saudi Arabia. The GCC states and Jordan appear at the top of the list.

WF4.036 — The number field accepts spaces and dashes and normalises them. Leading zeros after the country code are stripped: 0501234567 with +966 selected must succeed.

WF4.037 — Terms and Privacy links open in-app, in the user's selected language. The checkbox is unticked by default and Send code is disabled until it is ticked.

WF4.038 — Six-digit code, valid for 10 minutes. Auto-read from SMS on Android (SMS Retriever API) and iOS autofill. Auto-submits at six digits.

WF4.039 — Resend available after 45 seconds. Maximum 5 code requests per identifier per hour, enforced server-side.

WF4.040 — After 5 incorrect code entries the account is locked for 15 minutes and the user is shown the Wafra contact options (F13).

4.8 A7 — Your details and units

←

Tell us about you

Your name *

Set a password *

.....

⓪

At least 8 characters

How do you measure land? *

Dunum

Hectare

Acre

We have set this from your country.
You can change it in Settings.

Continue

WF4.041 — Name is mandatory. Password minimum 8 characters, no composition rules.

WF4.042 — A show/hide control on the password field.

WF4.043 — The **land unit is asked here**, pre-selected from the country detected on the phone: dunum in the UAE, hectares in Saudi Arabia, acres elsewhere. Changeable in Settings at any time (§10.4).

WF4.044 — Password is optional for Worker accounts created by invitation; those may sign in by code alone. Owners and Supervisors must set one.

WF4.045 — A user who reaches A7 through *Create an account* becomes the **Owner** of the first farm they create. There is no separate screen asking how they intend to use the app — A2 already established that.

WF4.046 — Around 80% of accounts are farm owners and 20% supervisors. Supervisors arrive by invitation (A15), not through this path.

4.9 A8 — What do you grow?

←
What do you grow?
You can change this later

*
Field crops
Wheat, alfalfa, potato,
tomato

♣
Date palms and fruit trees
Date palm, citrus, mango,
grape

* ♣
Both

WF4.047 — Three options : Field crops, Date palms and fruit trees, Both. This sets the default farm type for the first farm and determines which service is offered at A13. It does not restrict the account.

WF4.048 — The tree category is called **date palms and fruit trees** throughout the app. "Orchard" is not the local term and is not used.

WF4.049 — The examples under each option are **all at the same level — individual crops**. Not a mix of crops and groupings: "wheat, alfalfa, potato, tomato", never "alfalfa ... fodder", which names the same thing twice at two levels. See the open items for the wider nomenclature question.

WF4.050 — *Both* carries no example line. Its meaning is complete without one.

4.10 A9 — Add your first farm

Two ways in. The farmer chooses once, per farm.

▸ Search a place, address or
coordinates

Add your farm

■ Survey my whole farm
Draw the outer boundary and we
find the plots and the trees

📍 Draw my plots myself
Trace each plot and tell us
what is growing in it

You can add more plots or run a
survey later.

WF4.051 — This screen is reachable only by Owners and Supervisors with the `plot.create` capability. It never appears for a Worker.

WF4.052 — Both routes are offered with equal weight. Neither is presented as the advanced or the expert option.

WF4.053 — The choice is per farm, not per account. A farmer may draw one farm by hand and survey the next.

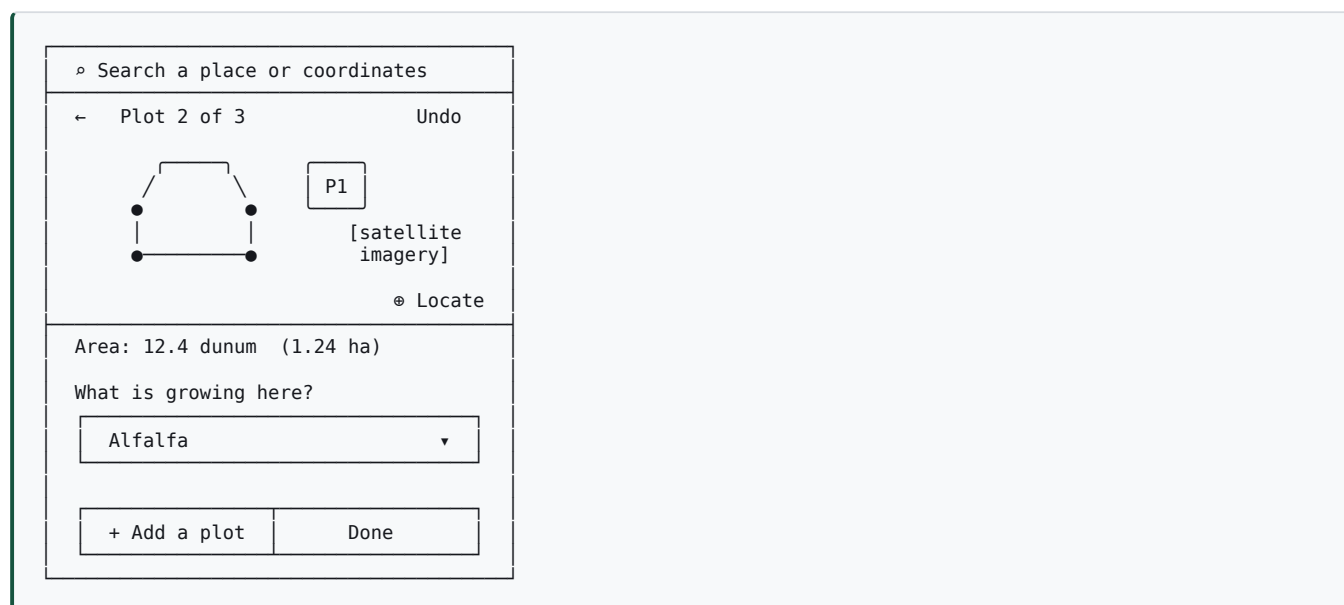
WF4.054 — The survey route is offered again from Farm settings (B11) for a farm that was created by hand, and *Add a plot by hand* remains available on a surveyed farm.

WF4.055 — The app opens the satellite map **centred on the device's location** and lets the farmer scroll to their land.

WF4.056 — A **search bar is present and visible at all times** on every map screen. It is not behind an icon, a drawer or a control. It accepts a place name, a community name, an address or a pair of coordinates, and it also accepts a dropped pin.

WF4.057 — Search is served by the Google Maps API.

4.10.1 A9 — DRAW MY PLOTS MYSELF



WF4.058 — **The manual route supports many plots per farm.** The farmer draws a plot, names what is growing in it, and adds another. A farm is not limited to one hand-drawn boundary.

WF4.059 — Each hand-drawn plot carries its own crop entry. Plots on the same farm may hold different crops.

WF4.060 — There is **no limit on the number of points** in a boundary.

WF4.061 — Where the farm holds trees, the survey still runs on the drawn plots, because the tree count determines the price (§4.10.4). The farmer is told this before they finish.

WF4.062 — Boundary auto-detection from a single tap in the middle of a field is not offered. The farmer either draws, or asks for a survey.

WF4.063 — Opens centred on the device's GPS position at approximately 1:5,000. If location permission is refused, opens on the search box.

WF4.064 — Satellite basemap by default.

WF4.065 — Live area readout in the user's chosen unit, with hectares in brackets. See §10.4.

WF4.066 — *Undo* removes the last placed vertex. *Clear all* is available behind a confirmation.

WF4.067 — Vertices are draggable with a touch target of at least 48 dp, offset above the fingertip so the dragged point is not hidden by the finger.

WF4.068 — Self-intersecting polygons are rejected with a plain-language message and the offending segment highlighted.

WF4.069 — A polygon smaller than 0.1 ha or larger than 10,000 ha triggers a confirmation.

4.10.2 A10 — SURVEY MY WHOLE FARM

The farmer draws one polygon around everything they hold. MMC's land use algorithm then classifies what is inside it.

Search a place or coordinates

← Your farm boundary Undo

[satellite imagery]

⊙ Locate

Area: 240 dunum (24 ha)

Draw around everything you hold – fields, trees, buildings and all. We work out what is farmed, and what your subscription will cost.

Send for survey

Usually ready in 15–20 minutes

WF4.070 — The drawing interaction is the one at A9 : same vertex handling, same undo, same area readout, same validation (WF4.063, WF4.067, WF4.068, WF4.069).

WF4.071 — One polygon per survey. Holdings that are not contiguous are surveyed as separate farms.

WF4.072 — On *Send for survey* the farm record is created immediately in a **surveying** state and the farmer is returned to Home. The app does not hold them on a progress screen.

WF4.073 — The expected wait is stated on the button and on the farm card: **15 to 20 minutes**. The figure is server configuration, not a compiled-in constant.

WF4.074 — A survey can be requested without an active subscription and without a payment method on file. Nothing is charged before the farmer accepts a plan at A13.

WF4.075 — Only one survey may be in flight for a given farm. While one is running the action is disabled and labelled *Survey in progress*.

WF4.076 — A survey that has not returned within twice the stated time shows a plain notice on the farm card with a *Contact Wafra* action (F13). It never sits silent.

WF4.077 — A survey that fails offers two actions : try again, or draw the plots by hand (A9). It never leaves the farmer with a farm that cannot be priced.

4.10.3 A11 — WHAT WE FOUND

Returned when the survey completes. Colour-coded map, then the same areas as an editable list. **The result is a starting point, not a verdict.**

← What we found

[classified overlay]

Field crops Trees Excluded

✓ P1	Field crops	44 dunum	Edit ▾
✓ P2	Field crops	44 dunum	Edit ▾
✓ P3	Date palms	64 dunum	Edit ▾
	1,180 trees		Edit ▾
✗	Buildings & covered	12 dunum	Edit ▾

Split Join Remove + Add plot

Included: 152 dunum · 1,180 trees

Confirm and continue

WF4.078 — Three classes are returned: **open field crops** including fallow land, **date palms and fruit trees**, and **covered agriculture and structures** — greenhouses, villas, housing, stores and yards.

WF4.079 — Colour on the map is never the only signal. Every area also appears in the list below with its class in words, its area and its state (WF2.008).

WF4.080 — Covered agriculture and structures default to excluded. The other two classes default to included.

WF4.081 — The farmer can **split** a plot the survey merged, **join** plots it separated, **edit** any boundary, **remove** a plot and **add** a plot the survey missed. Splitting and joining are the answer to adjacent parcels the algorithm read as one.

WF4.082 — Boundary editing here uses the same interaction as A9 and the boundary editor (C5) : tap to add a point, drag to move, delete the selected point.

WF4.083 — Every area can be included or excluded, and its class corrected — field crops to trees, structures to field crops, and so on.

WF4.084 — Totals update as the farmer changes anything: included area, crop area, tree count, excluded area, each in the user's unit with hectares in brackets.

WF4.085 — On *Confirm and continue*, every included area becomes a plot, carrying its class as the plot type. Excluded areas are stored with the farm and are not analysed, not counted and not charged for.

WF4.086 — The classification the algorithm returned and the farmer's corrections are both retained. A corrected area records what was detected, what it was changed to, and by whom.

WF4.087 — A survey that returns no farmed area at all shows a plain explanation and two actions : draw the plots by hand (A9), or contact Wafra (F13).

WF4.088 — The farm stays in the **surveying** state until this screen is confirmed. An unconfirmed result is re-openable from the farm card for 30 days.

4.10.4 FROM SURVEY TO PLAN

WF4.089 — After A11 the farmer goes to A12 (farm details) and then to A13, where the service and the level are set : crops only, trees only, or both, and basic or pro within that (§9).

WF4.090 — The price is calculated from what the survey found. Crop capability is priced per unit of included crop area; tree capability is priced per included tree. There is no static price list in the app.

WF4.091 — No price is shown before the survey has completed and been confirmed. A13 is not reachable from a farm still in the surveying state.

WF4.092 — Where the farmer changes the included areas at A11, the price at A13 recalculates from the new totals before it is shown.

WF4.093 — A13 is pre-selected from what the survey found — crops only, trees only or both — and the farmer can change it.

WF4.094 — Changing the service at A13 does not discard the excluded or unselected areas. A farmer who buys crops only keeps the tree areas on record and can add them later without a second survey.

4.11 A12 — Farm details

← Farm details

Farm name *

Al Kharj North

What is on this land? *

() Field crops

(●) Date palms and fruit trees

() Both

Soil type

Sandy loam ▼

Not sure? We will estimate it

Save farm

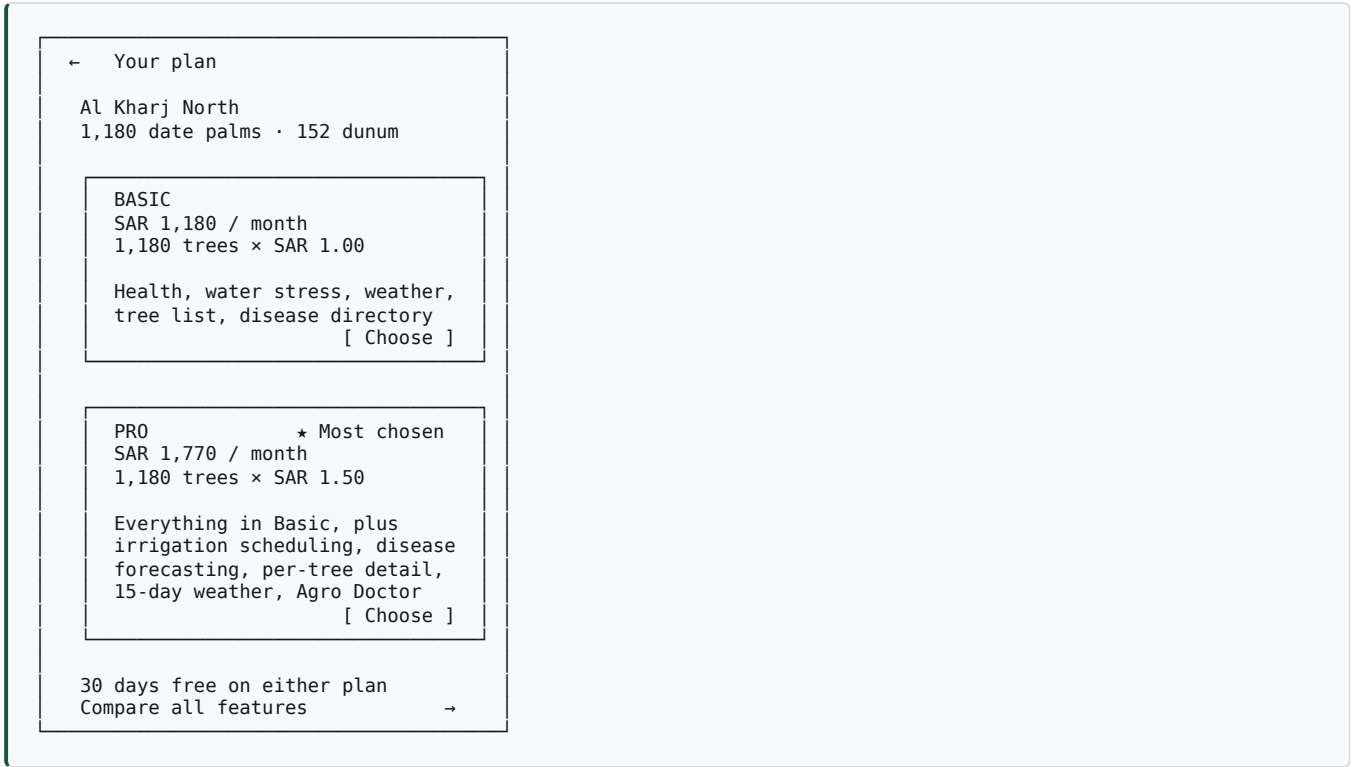
WF4.095 — Only farm name and farm type are mandatory. Every other field is optional and can be completed later.

WF4.096 — The irrigation system is not collected. Drip, sprinkler, pivot, bubbler and mixed systems are out of scope for this release — not asked at setup, not stored, not shown. Irrigation efficiency and pump flow rate are separate figures and are still used, as editable assumptions on the advice screen (§5.8.2).

WF4.097 — Soil type may be left blank; the backend fills it from the soil layer where available and the app shows it as an estimate the user can override.

4.12 A13 — Your plan and price

WF4.098 — The plan chooser shows only the service matching the farm just created, and the price it computes from that farm's survey.



WF4.099 — The plan cards carry a **live calculation**, not a static price: the quantity the survey found, the rate, and the resulting monthly figure, shown as an arithmetic the farmer can follow.

WF4.100 — Both a monthly and an annual figure are available on each card.

WF4.101 — Two levels per service, always in the order **Basic** → **Pro**.

WF4.102 — Rates are fetched from the server, never hard-coded, and displayed in the currency of the user's country. See §10.4.

WF4.103 — *Compare all features* opens the full comparison table (F6).

WF4.104 — Adding a **sixth farm** shows an enterprise screen inviting the farmer to talk to an advisor, alongside the option to continue on the standard plan.

WF4.105 — An account holds a maximum of **10 farms**. At the cap the *Add farm* action explains the limit and offers the enterprise contact route.

4.12.1 WHEN A CLIENT HAS BOTH CROPS AND TREES

Crop capability is priced by area and tree capability by tree count. A client holding both must not be asked to buy and manage two subscriptions.

WF4.106 — Where the account has at least one crop farm and at least one tree farm, the plan chooser offers a **combined** service and does not offer the two separately:

Combined	Grants
Combined Basic	Crop Basic on every crop area, Tree Basic on every tree
Combined Pro	Crop Pro on every crop area, Tree Pro on every tree

Open Item : is a combined holding sold as one plan at a reduced rate, or as two plans bought separately?
The screens above assume one plan. If it becomes two, A13 and F5 both change.

WF4.107 — A combined plan is a **single purchasable product** with a single price and a single renewal date. The app never presents it as two line items.

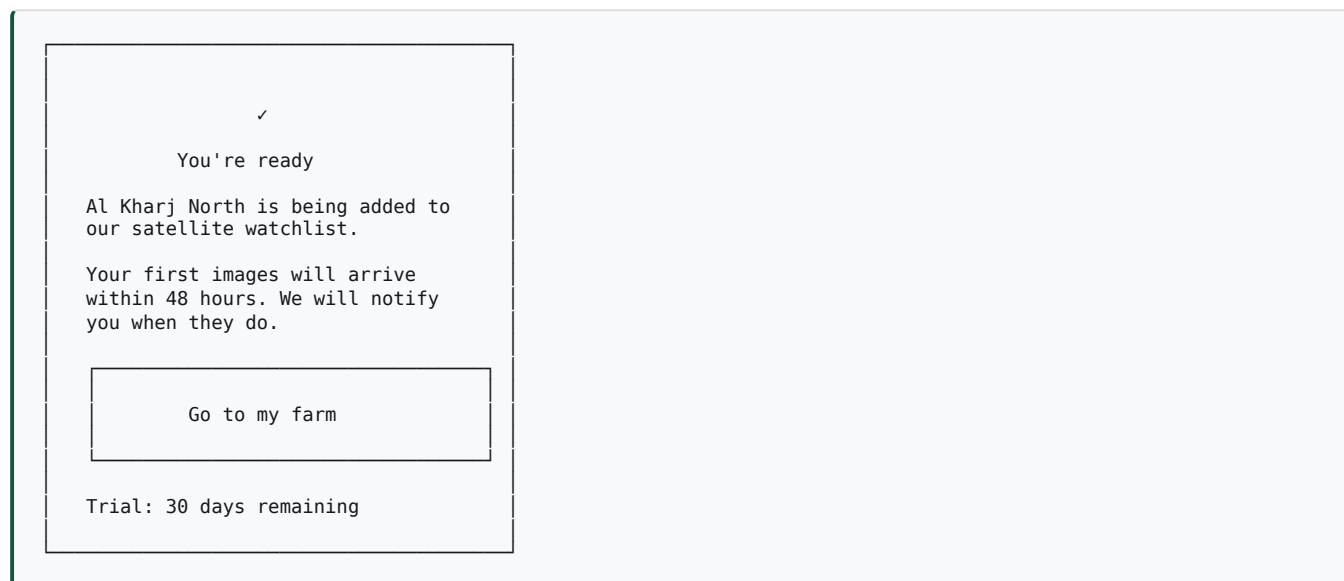
WF4.108 — A mixed farm — crops and trees inside the same farm boundary — requires a combined plan. The app states this at A12 when *Both* is selected as the farm type.

WF4.109 — Where a client on a crop-only plan adds their first tree farm, the app offers an upgrade to the matching combined plan rather than a second subscription. The reverse applies when a tree-only client adds a crop farm. Until they upgrade, the new farm is visible but its analytics are locked, with the upgrade path shown in place of the data.

WF4.110 — Entitlement is resolved server-side as a single set of feature keys. The app asks what the user may do and never derives capability from the plan name or the farm type (\$9.5).

WF4.111 — Downgrading from a combined plan to a single service is only possible once the farms of the other type have been archived. The app explains which farms block the change and offers to archive them.

4.13 A14 — You're ready



WF4.112 — A confirmation screen stating when the first satellite imagery will be available, and a single button to Home.

4.14 A15 — Join a farm

← Join a farm

Enter the code you were sent

K 7 M

■ Scan QR code

Join

No code? Ask the farm owner to send you one.

WF4.113 — Opening an invitation deep link, with or without the app installed, leads to: install if needed → verify mobile number → land inside the farm they were invited to, with their role already applied.

WF4.114 — An invited user never sees A8–A13. They join an existing subscription.

WF4.115 — Invitations expire after 7 days and are single use. See §4.1.

WF4.116 — A used, expired or revoked invitation shows a clear message and a *Contact the farm owner* action. It never fails silently and never grants partial access.

WF4.117 — On successful join, a Worker lands directly on My Work; a Supervisor lands on Home filtered to the farms they were granted.

5 Screen specifications

5.1 B1 — Home / My farms

The first screen after login for Owners and Supervisors.



WF5.001 — The summary card at the top counts farms by worst-plot status, not by average. A farm with nine healthy plots and one urgent plot is an urgent farm.

WF5.002 — *See what to do* navigates to the Advice inbox (D1) pre-filtered to Needs attention and Urgent.

WF5.003 — Each farm card shows: status icon and colour, farm name, farm type icon, area, plot or tree count, a one-line plain-language summary of the most significant issue, and the age of the data.

WF5.004 — "Updated X ago" refers to the age of the most recent **satellite imagery**, not the last app refresh. Where imagery is older than 7 days, show the reason ("Cloud cover since 28 July").

WF5.005 — A farm awaiting its survey result shows in place of a status card: "Survey in progress — we are working out what is growing on your farm." A farm whose result is back and unconfirmed shows "Survey ready — confirm what we found" and opens A11.

WF5.006 — A farm in either survey state is not counted in the summary card at the top and carries no health status.

WF5.007 — **All farms** is a first-class view. The farmer can see every plot across every farm in one list and on one map, as well as one farm at a time. Moving between the two never requires going back a screen.

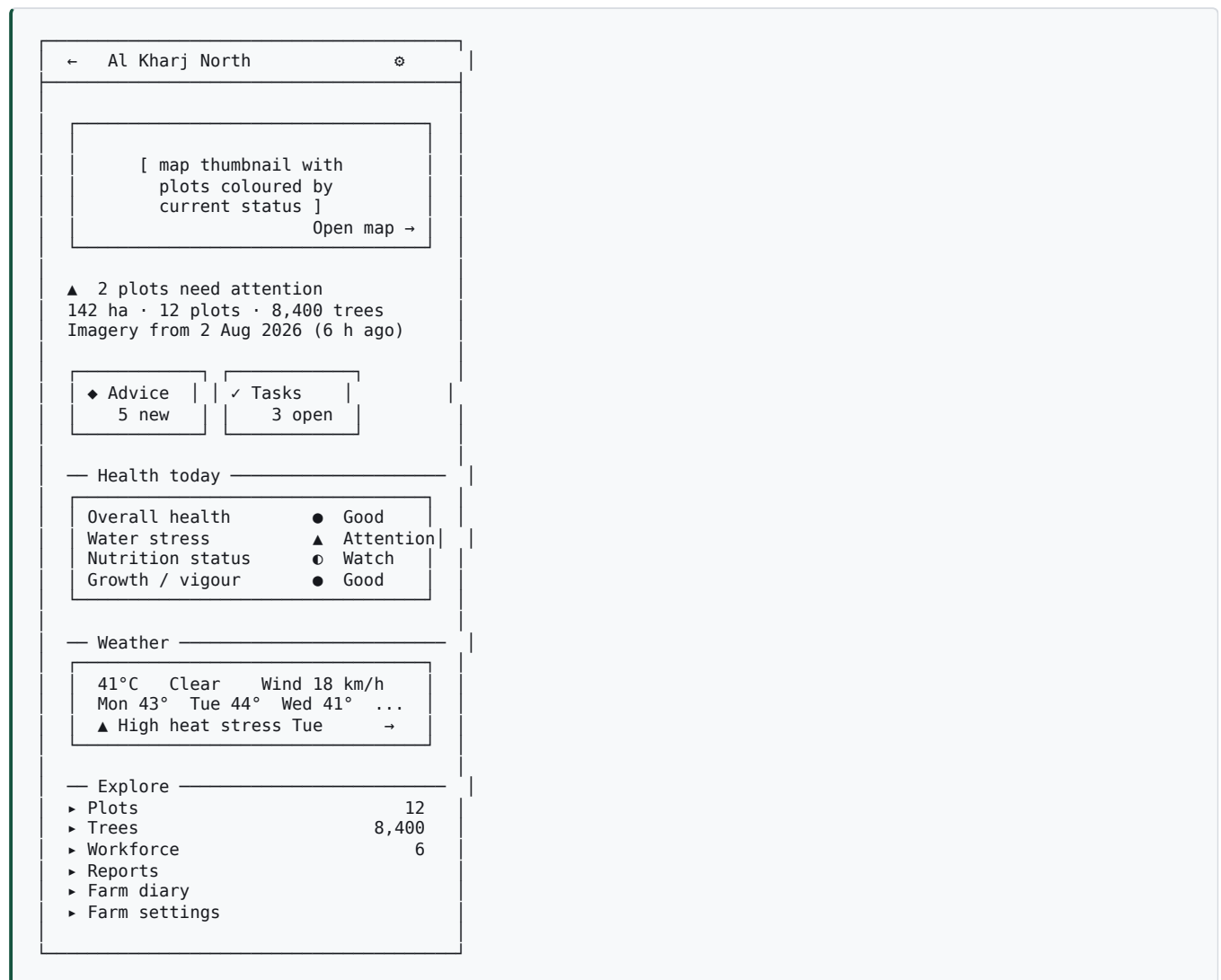
WF5.008 — The map toggle icon that appears beside each individual farm also appears beside the **ALL FARMS** row, and opens the map with every farm's plots on it.

WF5.009 — Farms are sorted by severity first, then by most recently viewed.

WF5.010 — Pull to refresh, with a visible refresh button as the non-gesture equivalent.

WF5.011 — Empty state: "You have no farms yet" with an *Add your first farm* button leading to A9.

5.2 B2 — Farm detail



WF5.012 — The health block lists only measures included in the farm's current plan. Measures outside the plan are shown greyed with a small lock and a tap target that opens the upgrade screen (§9.5) — never hidden entirely. The user should know what they are missing.

WF5.013 — Tapping any row in the health block opens the measure viewer (B7) for that measure across the whole farm.

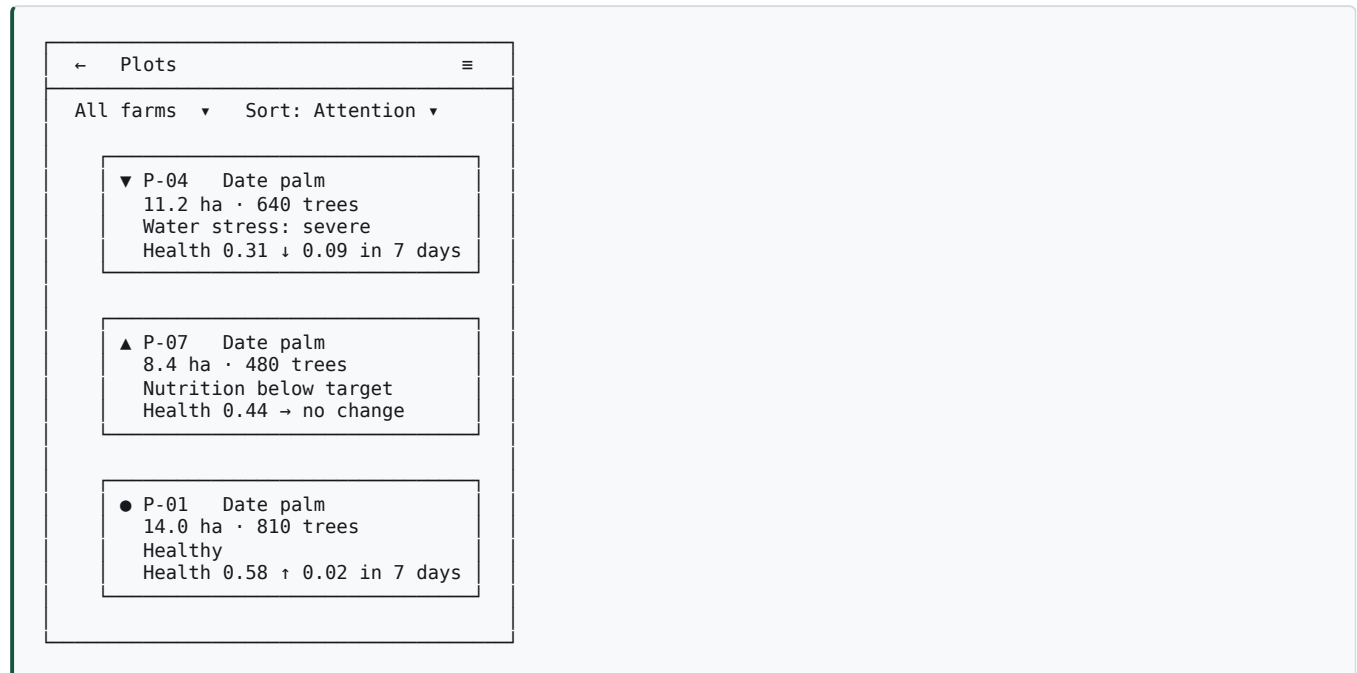
WF5.014 — The *Trees* row is present only for tree and mixed farms.

WF5.015 — The weather strip shows current conditions and a 7-day or 14-day forecast according to plan (§9.3), and surfaces any active weather alert inline.

WF5.016 — Farm settings (⚙, top right) is visible to Owner and Supervisor only.

WF5.017 — Workforce sits on the farm screen, in the Explore list beside Plots and Trees, showing the number of people on this farm. It is not buried in Settings or the More tab (§5.6).

5.3 B3 — Plots



WF5.018 — Plots are listed flat under the farm. **There is no grouping by block.** Blocks and fields do not exist in the schema and must not appear in any list, map layer or filter.

WF5.019 — The list has a farm selector at the top with **All farms** as one of its options, so plots can be read across the whole account or one farm at a time.

WF5.020 — Sort options: Needs attention first (default), Name, Area, Most recently changed.

WF5.021 — Each plot row shows the measure value and its change over the last 7 days, with a direction arrow.

5.4 B4 — Plot detail



WF5.022 — The measure selector is a dropdown listing every measure available on this farm's plan, in plain-language names with the technical name secondary:

Shown to user	Technical
Overall health	NDVI
Water stress	NDWI / water-stress measure
Nutrition status	NDRE / chlorophyll
Growth and vigour	EVI
Soil-corrected health	MSAVI
Photosynthesis	as provided by MMC

The plain-language name is what appears everywhere in the interface. The technical name appears in the measure viewer legend and in reports, for the agronomists.

WF5.023 — Date stepping with ◀▶ moves between **available imagery dates**, not calendar days. Dates with no imagery are skipped, and the reason is shown if the user reaches the end of available data.

WF5.024 — The interpretation line below the map names where and how long, not only what.

WF5.025 — *See what to do* is the primary action and links to the relevant recommendation (D2–D4, D6). Where no recommendation exists for this plot, the button reads *Nothing to do here today* and is inert. **There is no create-task action on this screen** (§5.8.1).

WF5.026 — The  menu contains: Edit plot, Edit boundary, Add crop cycle, Add observation, Share plot summary, Delete plot. Delete requires typing the plot name to confirm and is Owner-only.

WF5.027 — Plot detail aggregates the whole plot: area, tree count and a severity per measure. **It does not split figures by cultivar or variety.**

WF5.028 — Years are rendered as plain four-digit numbers with no thousands separator. "1990", never "1,990". This applies everywhere a year appears.

5.4.1 B7 — MEASURE VIEWER, FULL SCREEN

WF5.029 — Full-screen measure map, pinch-zoom and pan, with a persistent legend showing the value scale and the units.

WF5.030 — Tapping any point on the map shows the value at that point in a small callout, with its coordinates.

WF5.031 — The legend scale is fixed per measure, not auto-scaled to the current image, so that week-to-week comparison is valid.

5.4.2 B8 — COMPARE

WF5.032 — Side-by-side or swipe-divider comparison of the same plot on two dates, with both dates labelled and the difference in the measure value stated numerically.

WF5.033 — A *compare with previous years* mode showing the same calendar week across up to 5 previous years. Pro only; on Basic the control is visible and locked.

5.4.3 B5/B6 — CROP CYCLES

A crop cycle is a first-class record. It is **not** a field on the plot.



WF5.034 — Adding a new crop cycle to a plot that already has an open one prompts the user to close the existing cycle first, recording a harvest date and — optionally — a yield. It must not silently overwrite.

WF5.035 — Closing a crop cycle never deletes it. The plot's full crop history is retained indefinitely and remains visible. Rotating a field must not erase multi-year history.

WF5.036 — Crop cycle fields: crop, variety, sowing/planting date, expected harvest date, actual harvest date, target yield, actual yield, notes. Only crop and start date are mandatory. Irrigation method is not among them (WF4.096).

WF5.037 — Yield here is **what the farmer recorded**, not a prediction. The app does not estimate yield (§9).

WF5.038 — The crop picker is searchable, grouped by category (cereals, forage, vegetables, fruit trees, other), and offers the last five crops used on this account at the top.

5.4.4 MULTI-CROP AND HIERARCHY

WF5.039 — A farm supports many plots with different crops, varieties and planting dates simultaneously.

WF5.040 — The hierarchy is **farm** → **farm boundary** → **plot** → **tree**. There is no block and no field, in the data model or in the interface.

WF5.041 — Plots sit directly under the farm boundary. A plot may be split into two, or two plots joined into one, at any time — from A11 at survey confirmation and from the boundary editor (C5) afterwards. Splitting and joining preserve the history of both sides.

5.4.5 INTERCROPPING

WF5.042 — A plot may record a primary crop and one or more secondary crops (for example date palm with alfalfa beneath).

WF5.043 — Measure values and health status on an intercropped plot are attributed per crop. The plot detail screen carries a crop selector alongside the measure selector, defaulting to the primary crop; a Whole plot option shows the unseparated reading.

WF5.044 — Recommendations are generated per crop within the plot. An irrigation or nutrition recommendation names the crop it applies to, and the task created from it carries that crop.

WF5.045 — Where separation cannot be computed for a given date (e.g. insufficient resolution, cloud, or an unsupported crop pairing) the app shows the whole-plot reading marked Combined canopy and suppresses per-crop recommendations for that date only. It never presents a whole-plot figure as if it were a single crop's.

5.4.6 B11 — FARM SETTINGS

WF5.046 — Editable: farm name, farm type, address/region, soil type, default language for reports, primary contact. Owner and Supervisor only. Irrigation system is not collected (WF4.096).

WF5.047 — Farm settings offers *Run a survey on this farm* for a farm created by hand, and *Add a plot by hand* for a farm created by survey.

WF5.048 — Contains *Transfer ownership* (Owner only, with SMS confirmation) and *Delete farm* (Owner only, requires typing the farm name, warns that all history is lost, and requires a second confirmation after 24 hours for farms with more than 90 days of data).

5.4.7 B12 — ADD FARM

WF5.049 — *Add farm* runs the same sequence as A9: draw the plots, or send a boundary for survey. It never bypasses the survey for a farm holding trees, because the tree count sets the price (§9.1.1).

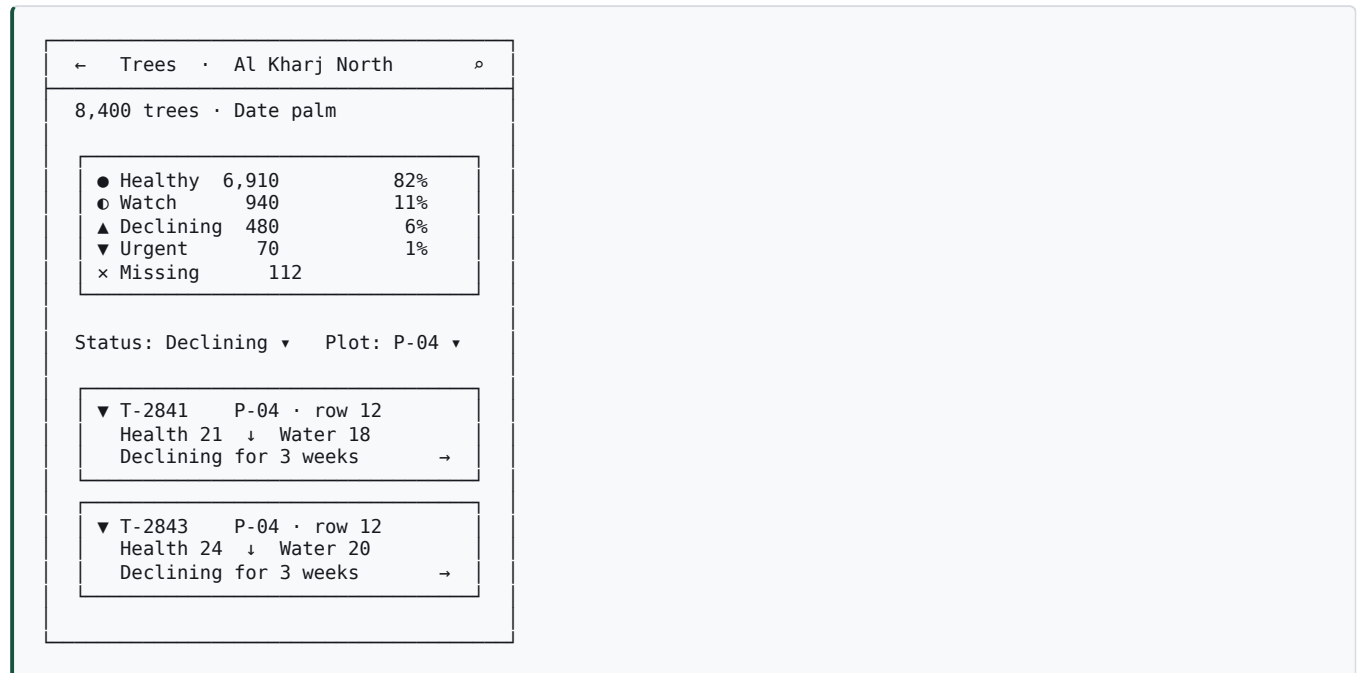
WF5.050 — Adding a **sixth** farm shows the enterprise screen first (WF4.104), with the option to continue on the standard plan.

WF5.051 — Adding an **eleventh** farm is refused with a plain explanation and the enterprise contact route (WF4.105).

WF5.052 — A farm added to an account of the other type follows WF4.109: visible, boundary kept, analytics locked behind the upgrade to Combined.

5.5 B9/B10 — Trees

Shown for tree and mixed farms only.



WF5.053 — The screen leads with the status distribution; the list below is filtered to what needs action by default.

WF5.054 — Filters are **status** (healthy, declining, missing) and **plot**. Fruit type appears as a filter only where the farm actually holds more than one type — a pear filter is never offered on a farm with no pear trees.

WF5.055 — **The tree list creates no tasks.** Filtering to a condition — four declining trees on P-02, seventy showing the same stress — is an analytics view. Where those trees need work, the advisory layer raises it and the task is created there (§5.8.1).

WF5.056 — Tree detail (B10) shows: tree ID, **tree type**, its plot and row, a locator view (§5.7.1), the current health score and its component measures, a 12-month health trend chart, and the history of observations, tasks and photos attached to that tree.

WF5.057 — **Planting year is not carried on the tree record.** The tree record carries its type instead.

WF5.058 — Tree detail has **no create-task action** (§5.8.1).

WF5.059 — Missing and dead trees are shown as a distinct state with their own count, not folded into "urgent".

WF5.060 — Where a tree-level measure is not in the client's plan, it is listed and locked rather than omitted (WF5.012).

WF5.061 — Analysis is per tree and per plot. Every canopy carries its own profile and its own tree ID.

5.6 G1–G3 — Workforce

Most farm labourers will never install the app. Many cannot read. A farm typically has two or three of them, and the owner is the one holding the phone. The app therefore keeps **worker records** the owner can assign work to directly.

The screenshot shows a mobile app interface for the 'Workforce' section. At the top, there is a header bar with a back arrow, the title 'Workforce', and a '+ Add' button. Below the header, the section is titled 'Al Kharj North'. There are two worker cards listed:

- AR Ahmed Rahman**: +880 1X XXX XXXX, Hindi · WhatsApp, 3 open tasks. A right arrow is visible next to the WhatsApp indicator.
- BH Bilal Hossain**: +966 5X XXX XXXX, Bengali · SMS + WhatsApp, 1 open task. A right arrow is visible next to the WhatsApp indicator.

The screenshot shows the 'Add a worker' form. It has a header bar with a back arrow, the title 'Add a worker', and a 'Save' button. The form contains the following fields and options:

- Name ***: A text input field containing 'Ahmed Rahman'.
- Mobile number ***: A text input field with a country code dropdown set to '+880' and a number input field containing '1X XXX XXXX'.
- Language ***: A dropdown menu currently showing 'Hindi'.
- Work will be sent in this language**: A label text.
- How should we reach him?**: A section with two toggle switches: 'SMS' (set to 'off') and 'WhatsApp' (set to 'on').
- Save**: A large button at the bottom of the form.

WF5.062 — Workforce is a named feature with its own screens. Its home is the **farm screen** (B2), beside Plots and Trees (WF5.017). It is also reachable from the assignee picker. It is not in Settings and not in the More tab.

WF5.063 — A worker record carries exactly four fields: name, mobile number, language and notification preference. Nothing else is required to assign work.

WF5.064 — Creating a worker record does not create an app account and does not send an invitation. The person never has to install anything.

WF5.065 — Notification preference is per person, a yes/no toggle for **SMS** and a yes/no toggle for **WhatsApp**. Both may be on. At least one must be.

WF5.066 — Work sent to a worker record goes out as a plain message in that worker's language, carrying what to do, how much, where and by when.

WF5.067 — A worker record can be promoted to an app account later by sending an invitation to the same number (§4.1). The task history follows the person.

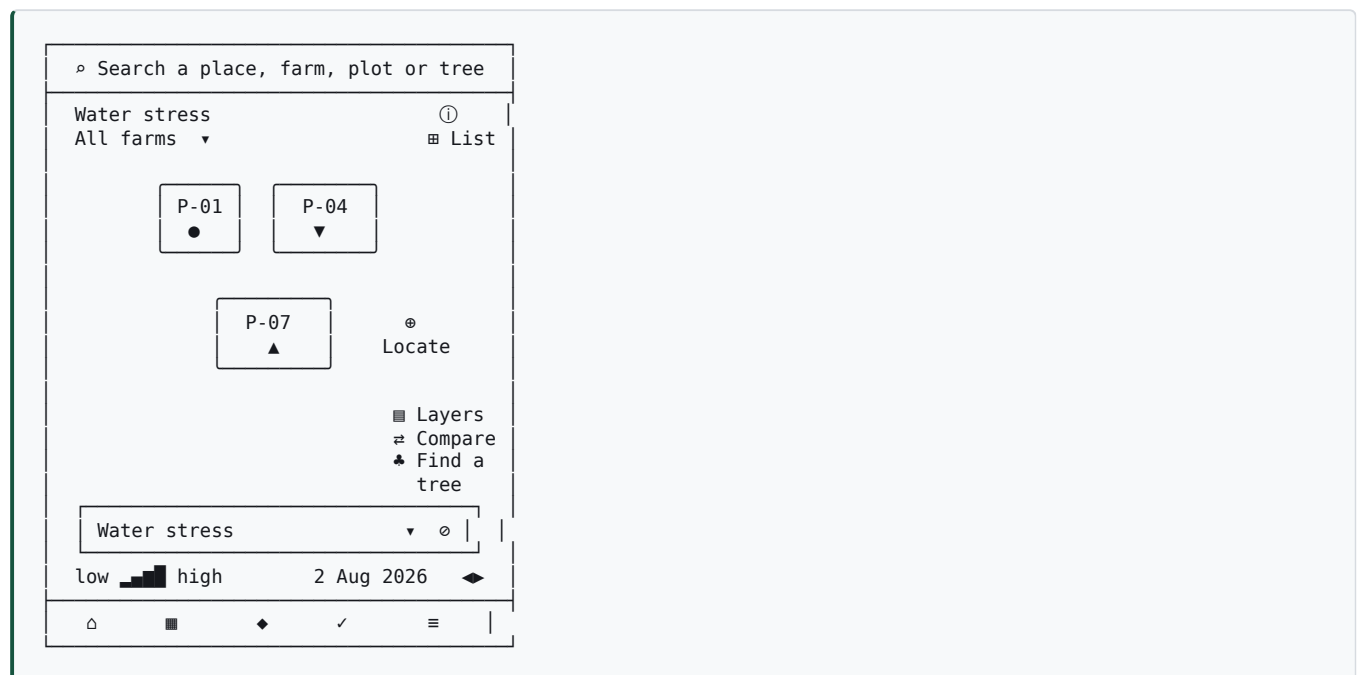
WF5.068 — Where a worker does have the app, their chosen app language overrides the language on the record and drives every message they receive.

WF5.069 — Workers who receive only SMS or WhatsApp never see a dashboard. Everything they need is in the message.

WF5.070 — A worker record can be deactivated. Their completed task history is retained and remains attributed to them.

Good to have, not required for launch : automatic translation of a task written by the owner into the recipient's language. Pre-packaged advisory tasks are already generated from the string catalogue in every language (§10.3); this covers free text the owner types themselves.

5.7 C1 — Map



WF5.071 — The map opens on the user's farms, zoomed to fit all of them, or to the currently selected farm where a farm context is active.

WF5.072 — Plot polygons are filled with the currently selected measure and carry their status icon and label at readable size. Labels hide automatically below a zoom threshold rather than overlapping.

WF5.073 — Tapping a plot opens a bottom sheet (C3) with: plot name, crop, area, current measure value and change, a status line, and one button — *Open plot* (B4). The sheet is draggable to full height without leaving the map. There is no create-task action here (§5.8.1).

WF5.074 — Layer picker (C2) offers, subject to plan:

- Basemaps: satellite, terrain, street
- Measure layers: all measures available on the plan
- Farm layers: **farm boundary, plot boundaries, plot labels, tree points**
- VRA layers: sowing, nitrogen, P&K, zoning (per plan)
- Irrigation map (Pro)

There is **no block layer** and **no field layer**. They do not exist in the schema.

WF5.075 — Layer selections persist between sessions.

WF5.076 — Locked layers appear in the list with a lock icon and open the upgrade screen when tapped.

WF5.077 — The user's own GPS position is shown when location permission is granted, with a *Locate me* control that centres on it.

WF5.078 — Date stepper on the map moves through available imagery dates for the visible plots, as WF5.023.

WF5.079 — Compare mode (C4) puts a draggable vertical divider across the map with a different date either side.

WF5.080 — Map tiles for the user's own farms are cached for offline use (§11.2). When offline, the map shows cached tiles and a clear banner stating the date of the cached imagery.

WF5.081 — The **search bar is visible at all times** on the map. It is not behind an icon or a drawer. It accepts a place name, a community name, an address, a coordinate pair, a farm name, a plot name, a crop name and a tree ID. Place search is served by the Google Maps API.

WF5.082 — The map header carries the **name of the measure currently displayed**. Its explanation sits behind an ⓘ button beside the name, never as text laid over the map.

WF5.083 — A **Find a tree** control sits on the map itself, because that is where the farmer is standing when they need it.

WF5.084 — The farm selector on the map offers **All farms** as well as each farm individually.

5.7.1 FINDING A TREE IN THE FIELD

Reached from the map (§5.7), from task detail (§5.9.2) and from tree detail (§5.5).

WF5.085 — *Show me where* opens the map centred on the target plot or tree, with a line and a distance from the user's current position. Not a routing engine.

WF5.086 — Identification in the field is by **GPS proximity and heading**. Standing beside a tree and pointing the phone at it surfaces that tree's record — nearest first, with the next two candidates listed beneath so the farmer can correct a wrong pick.

WF5.087 — Where GPS accuracy is worse than the tree spacing, the app says so plainly and falls back to the row: it names the row and the position along it, counted from the row end the farmer is nearest to.

WF5.088 — Tags and QR codes on trees are **deferred, not rejected**. They are workable on a small farm and unworkable at national scale. Nothing in the schema prevents adding a tag identifier later.

WF5.089 — The locator view works offline from cached tiles (§11.2) and states the age of the imagery it is drawn on.

5.7.2 C5 — BOUNDARY EDITOR

WF5.090 — Editing an existing boundary uses the same interaction as A9: tap to add, drag to move, and a visible delete control for the selected vertex. There is no limit on the number of points.

WF5.091 — The boundary editor also **splits one plot into two, joins two plots into one, removes a plot and adds a plot**, on any farm, at any time — not only at survey confirmation (§4.10.3).

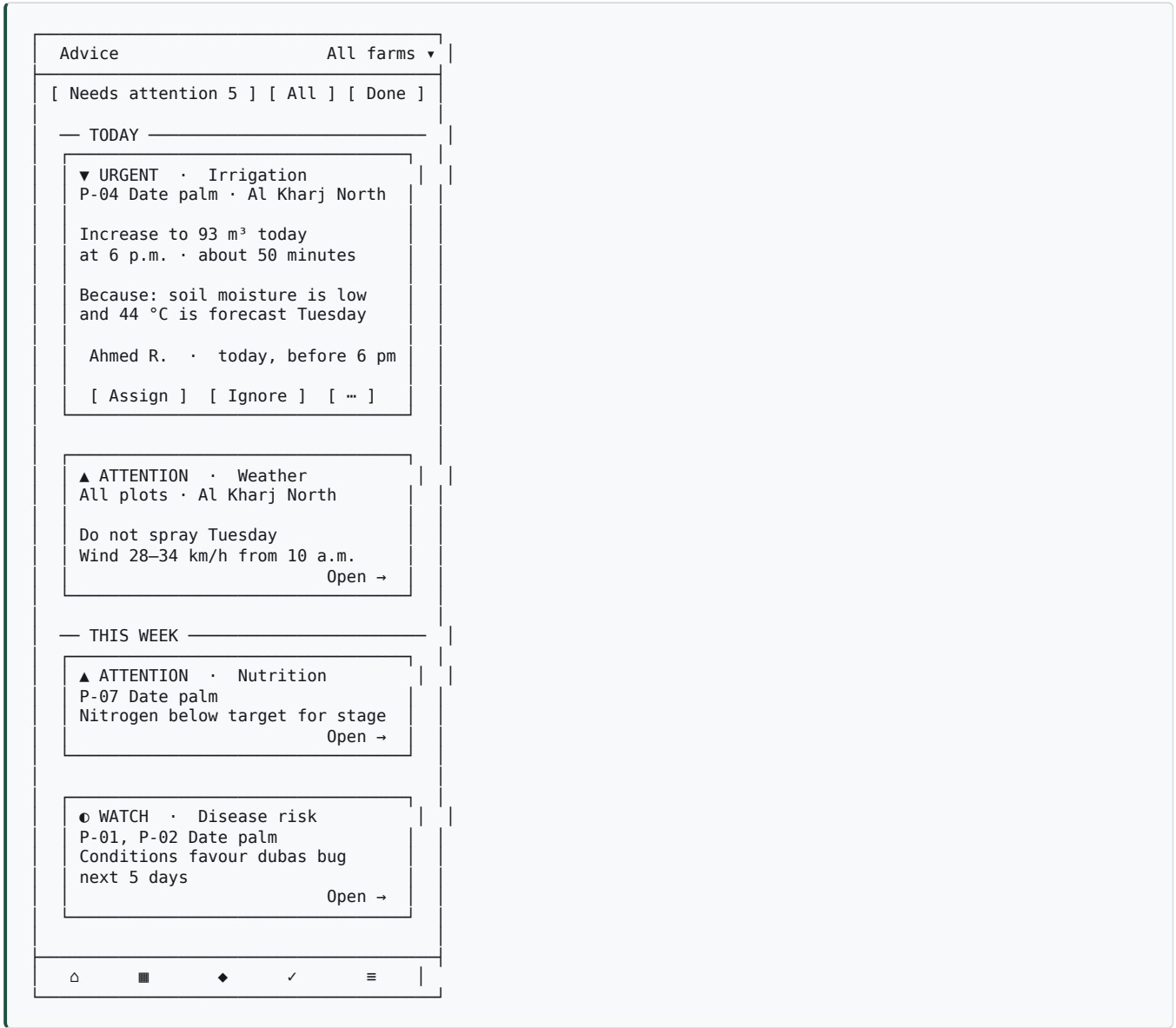
WF5.092 — A boundary change is a **versioned event**, not an overwrite. The previous geometry, the user who changed it and the timestamp are retained and appear in the activity log (F11). Historical analytics remain attached to the geometry that was in force at the time.

WF5.093 — Boundary editing requires the `farm.boundary.edit` capability (Owner and Supervisor). It is **not** available offline.

5.8 D1 — Advice inbox

This is the primary surface of the app. Everything else exists to support it. The system produces the advice and packages the work; the farmer assigns it, ignores it, or does it.

Functions like a message inbox : items arrive, are read, are acted on, and are cleared.



WF5.094 — Recommendations are grouped by urgency into Today / This week / Later, and within each group sorted by severity.

WF5.095 — Every recommendation card states, in this order and in plain language: what to do, how much, and why. The reason is mandatory.

WF5.096 — Every recommendation carries four actions, reachable without opening the detail screen:

Action	What it does
Assign	Sends the pre-packaged task to the named person and moves the item to Tasks
Ignore	Removes the item from the inbox until the next day
Remind me tomorrow	Same as ignore, stated as a deferral

Action	What it does
Mark as complete	Records that the work was done, without creating a task (D7)

WF5.097 — There is no interval menu on the reminder. The only option is **tomorrow**.

WF5.098 — An ignored item disappears from the inbox and returns the next day if the condition still holds. It is not deleted and it remains in the advisory log (§6.3).

WF5.099 — Every advisory item arrives **pre-packaged as a task**: description, timing and a suggested assignee already filled in. The farmer edits it, forwards it, or sends it as it stands.

WF5.100 — Before confirming, the farmer sees **what they are approving and why** — the instruction that will be sent, to whom, in which language, and the reason the system produced it.

WF5.101 — Tapping the body of a card opens the detail view: why it was suggested, what assumptions were made, and which measures were used. This is optional reading, never a step in the flow.

WF5.102 — Filters: farm (including **All farms**), plot, type (irrigation, nutrition, protection, weather), status (needs attention / all / done).

WF5.103 — A recommendation is marked done when the user records an action against it (D7) or completes a task created from it. It then moves to the Done tab and stops appearing in badges.

WF5.104 — Recommendations are never silently deleted or replaced. A superseded recommendation is marked superseded, with a link to the one that replaced it, and remains readable (§6.3).

WF5.105 — Where the client's plan does not include advisory, the Advice tab is still present and shows weather alerts (if in plan) plus a locked card describing what would appear here, with an upgrade path. The tab is never empty.

5.8.1 WHERE TASKS COME FROM

WF5.106 — **Advisory creates tasks. Task creation does not appear anywhere else** except the task screen itself. There is no *Create task* button on plot detail, tree detail, the map plot sheet, the farm screen, or any report.

WF5.107 — The task screen (E1) keeps an **ADD** button at the bottom right for the manual case. It is the **only** manual entry point in the app. There are no exceptions — not on plot detail, not on tree detail, not on the tree list, not on the map, not in reports.

WF5.108 — **Either an owner, a supervisor or a worker may add a manual task.** Manual task creation is not owner-only.

WF5.109 — A task may be completed by **the farm owner or the assigned worker**.

WF5.110 — Manual task creation is expected to be rare. It is built to work, not built to be prominent.

5.8.2 D2 — IRRIGATION ADVICE DETAIL

← Irrigation · P-04 ...

▼ URGENT · issued 3 Aug, 5:00 a.m.

INCREASE

93 m³ today
at 6:00 p.m.

About 50 minutes at 110 m³/h

Rest of the week

Wed 6 Aug	93 m ³	6:00 p.m.
Fri 8 Aug	88 m ³	6:00 p.m.

Why

- Crop water use today 8.4 mm
- Reference ET₀ 11.2 mm
- Crop coefficient 0.75
- No rain forecast this week
- Soil moisture at 40 cm: low
- Heat stress expected Tue (44 °C)

Assumptions we used Edit →
85% efficiency · sandy loam

Assign

Mark complete

This is advice, not a prescription.
Check conditions on the ground.

WF5.111 — Irrigation advice states **how much water the plot or the tree needs today**, not how much water it currently holds. Every irrigation item resolves to one of three recommendations: **increase**, **suspend**, or **maintain**.

WF5.112 — The recommendation carries **a volume and a time**: "93 m³ at 6 p.m.". A volume without a time is not a complete irrigation recommendation.

WF5.113 — Water is shown in **cubic metres only**. The parallel depth figure in millimetres is not shown to the farmer. Millimetres may still appear in the *Why* section as a model input.

WF5.114 — Times are shown on a **12-hour clock with a.m. and p.m.** The 24-hour clock is not used anywhere in the interface.

WF5.115 — Pumping time is calculated and displayed where the plot has a recorded system flow rate. Where it does not, show a prompt to add one rather than hiding the row.

WF5.116 — The "Why" section lists every input used, with its value.

WF5.117 — *Assumptions we used* is editable. A user who corrects their irrigation efficiency or soil type triggers a recalculation. The edit is recorded against the plot, not just this recommendation.

WF5.118 — The disclaimer line is present on every advisory detail screen, in the user's language, and is not dismissible.

5.8.3 D3 — NUTRITION ADVICE DETAIL

WF5.119 — Same structure: headline dose, split across application windows, the reasoning, editable assumptions.

WF5.120 — Nutrient recommendations are expressed **by element** — N, P, K, Ca, Mg and the rest — per unit area. Commercial product names are not used, because they differ from country to country. Where the user has recorded which products they hold, a translation is offered as a secondary line.

5.8.4 D4 — CROP PROTECTION ADVICE DETAIL

WF5.121 — Recommendations lead with the **active ingredient** and the rate, never a commercial product name.

WF5.122 — Product names are drawn from the registered-product register for the user's country — see §6.4.

WF5.123 — Every crop protection recommendation displays the **pre-harvest interval** for the user's country, prominently, with the earliest safe harvest date calculated and stated as a date, not as a number of days.

WF5.124 — A permanent, non-dismissible notice: *Check the product label and your local regulations before applying. This is advice, not a prescription.*

WF5.125 — The screen includes symptom photographs and a short identification guide from MMC's disease directory, so the user can confirm the diagnosis before spraying.

WF5.126 — Every crop protection recommendation is written to the advisory log when it is generated — see §6.3.

5.8.5 D6 — WEATHER ALERTS

WF5.127 — Alert types: frost, heat stress, high wind (spraying), heavy rain, sandstorm/dust, high humidity (disease). Each states the threshold crossed, the window, and one line on what it means for the farm.

WF5.128 — Weather alerts are delivered as push notifications at the criticality defined in §7.2, in the recipient's language.

5.8.6 D7 — RECORD WHAT YOU DID

← What did you do?

Irrigation · P-04 · advised 93 m³

✓
I applied the full amount

≠
I applied a different amount

×
I did not do this

Choosing a *different amount* reveals a single numeric field in cubic metres and an optional note. Choosing *I did not do this* reveals a short reason list (no water available, pump failure, weather, not needed, other).

WF5.129 — Recording an action takes at most **three taps** from the advice card for the common case.

WF5.130 — The recorded action writes to the plot's activity history and the farm diary, and is available to reports.

WF5.131 — Once actions have been recorded for a plot, the plot detail screen shows an advised-vs-applied comparison over time.

WF5.132 — This screen works offline.

5.9 E1 — Tasks

Not covered by the existing feature list. This module must be built.



WF5.133 — Task states: Open → In progress → Done, plus Cancelled. No other states, and no approval workflow.

WF5.134 — A task has: title, description, type, farm, plot or tree selection, assignee, due date and optional time, priority, and optional attachments.

WF5.135 — The **ADD** button sits at the **bottom right** of the task list and is the only manual task-creation control in the app (§5.8.1).

WF5.136 — On the task screen the **assigned person is at the very top**, the edit control at the bottom, and the ADD button bottom right.

WF5.137 — Task types (used for icons, filtering and reporting): irrigation, fertiliser, spraying, planting, pruning, harvest, inspection/scouting, maintenance, other.

WF5.138 — A task may be assigned to **one person**, who may be an app user or a §5.6 worker record. Team assignment is out of scope; a supervisor who needs three people creates three tasks or one task for a named lead.

WF5.139 — A task may cover many plots or many trees. Ten trees needing the same treatment are one task, not ten — the advisory item that raised them carries the whole group.

WF5.140 — Overdue tasks are always shown first, always visually distinct.

WF5.141 — **Scope: light task management, not a farm management system.** The module carries a task from advisory to a named person and back with evidence. It has no dependencies, no scheduling engine, no recurring tasks, no approval chain and no timesheets. Around nine in ten farmers are expected to use only what advisory generates.

WF5.142 — Where the platform can determine a schedule — irrigation timing, spray windows, harvest readiness — the app presents that determination rather than asking the farmer to enter it. Manual entry has been shown to confuse users in the field and is offered only where nothing can be derived.

5.9.1 E3 — CREATING A TASK

×

Send this task

Send

From: Irrigation advice · P-04

What

Irrigate plot P-04

Details

93 m³ at 6 p.m.
About 50 minutes at 110 m³/h

Type

↕ Irrigation ▼

Where

P-04 · Al Kharj North ▼

Who

Choose someone ▼

Ahmed R. Worker · Hindi
3 open tasks

Bilal H. Worker · Bengali
1 open task

Hassan M. Supervisor · Arabic
6 open tasks

When

Today

before 6 p.m.

Assign and send

Ahmed will receive this in Hindi

WF5.143 — The screen opens **fully pre-filled** when it comes from advisory, assignee included where the system can infer one. The farmer confirms, edits, or reassigns.

WF5.144 — The line under the save button names the language the assignee will receive the task in, before it is sent.

WF5.145 — Creating a task from a recommendation pre-fills title, description (including what and how much), type, plot and a suggested due date. The supervisor adds an assignee and saves.

WF5.146 — The assignee picker lists app users with access to the selected farm **and worker records held on the account** (§5.6), shows each person's role and language, and shows how many open tasks they already have.

WF5.147 — On save, the assignee is notified through their configured channel (§7) in **their** language, not the creator's.

5.9.2 E2/E4 — TASK DETAIL AND COMPLETION, WORKER VIEW

←

↑

Irrigate plot P-04

93 cubic metres

About 50 minutes

Today, before 6 p.m.

■ Show me where

✓ Mark as done

I cannot do this

WF5.148 — The Worker view of a task is stripped to: what, how much, where, when, and one big button. No status chips, no metadata, no comment threads.

Show me where opens the shared locator view — see §5.7.1.

WF5.149 — *Mark as done* opens E4. Completion may be recorded by the assigned worker **or by the farm owner** on their behalf (WF5.109):

← Mark as done

◆

Take a photo
(optional)

How much did you apply? (optional)

680

m³ ▼

Note (optional)

Confirm

● Offline – this will send when you have signal

WF5.150 — Everything on the completion screen is optional except the confirm tap. A worker who cannot type must still be able to complete a task.

WF5.151 — Photographs are automatically stamped with GPS coordinates, timestamp, task ID, plot ID and user ID. See §A.11.

WF5.152 — The whole completion flow works offline, confirms locally and immediately, and syncs later (§11).

WF5.153 — *I cannot do this* records a reason from a short picture-supported list and notifies the task creator immediately.

5.9.3 E6 — FIELD OBSERVATION / SCOUTING

× What did you see? Save

P-04 · Al Kharj North · located

Take a photo

photo1 +

What kind?

Pest Disease

Weed Water

Damage Other

How bad?

A few Patchy Wide-spread

Note (optional)

Save

● Offline – sends when you have signal

WF5.154 — Category and severity are chosen from labelled picture buttons, and only the photograph is required.

WF5.155 — Severity is expressed in plain words : a few, patchy, widespread.

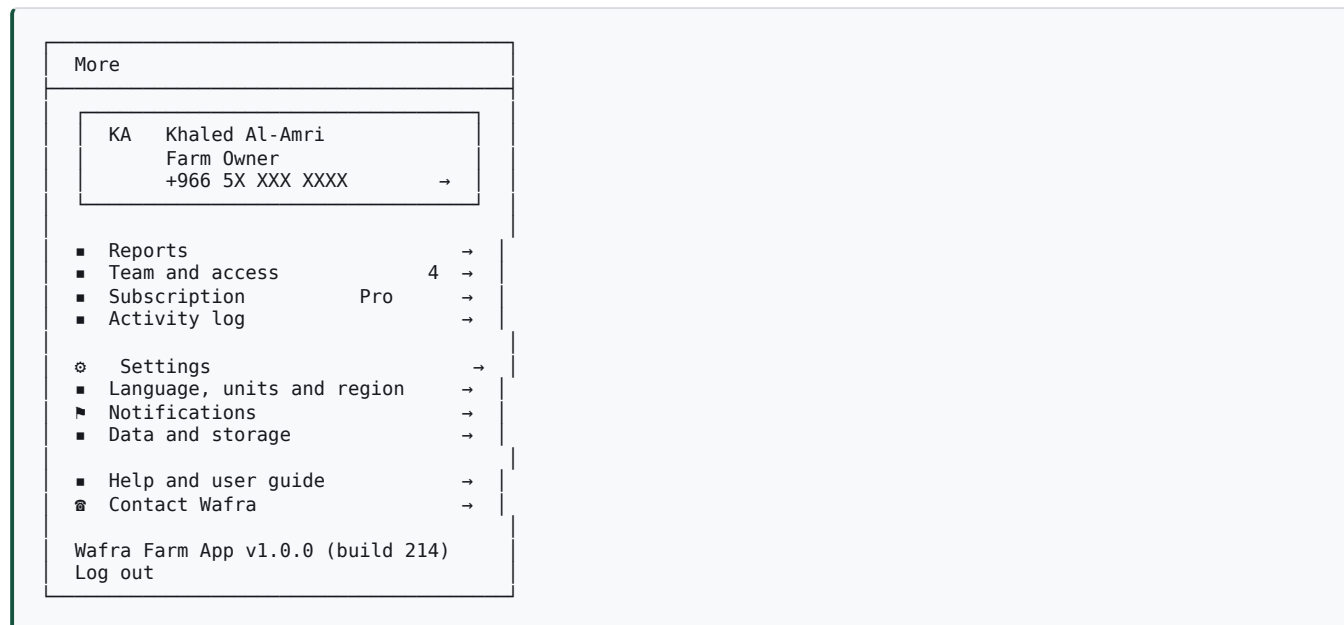
WF5.156 — A standalone observation can be captured from the Tasks tab, the map, a plot or a tree, without an existing task. Fields: photo(s), category (pest, disease, weed, water, damage, other), free note, severity, automatic location.

WF5.157 — Observations appear on the map as pins, on the plot's activity history, and in the farm diary.

WF5.158 — Where the plan includes photo-based disease detection (E7), photographing a leaf returns an automated identification with a confidence level, which the user may accept or reject. An automated identification is never presented as certain.

WF5.159 — Works fully offline. Photo disease detection requires a connection and is queued with a clear "will be checked when you have signal" state.

5.10 F — The More tab



WF5.160 — Menu contents are filtered by role. A Worker sees only: profile, language, notifications, data and storage, help, contact Wafra, version, log out.

WF5.161 — The version and build number are always visible on this screen.

5.10.1 F1 — REPORTS



WF5.162 — Reports are generated server-side as PDF. Excel export is offered for the tabular reports.

WF5.163 — Every report is produced in the language the user requests, from the same string catalogue as the app, and carries the Wafra brand.

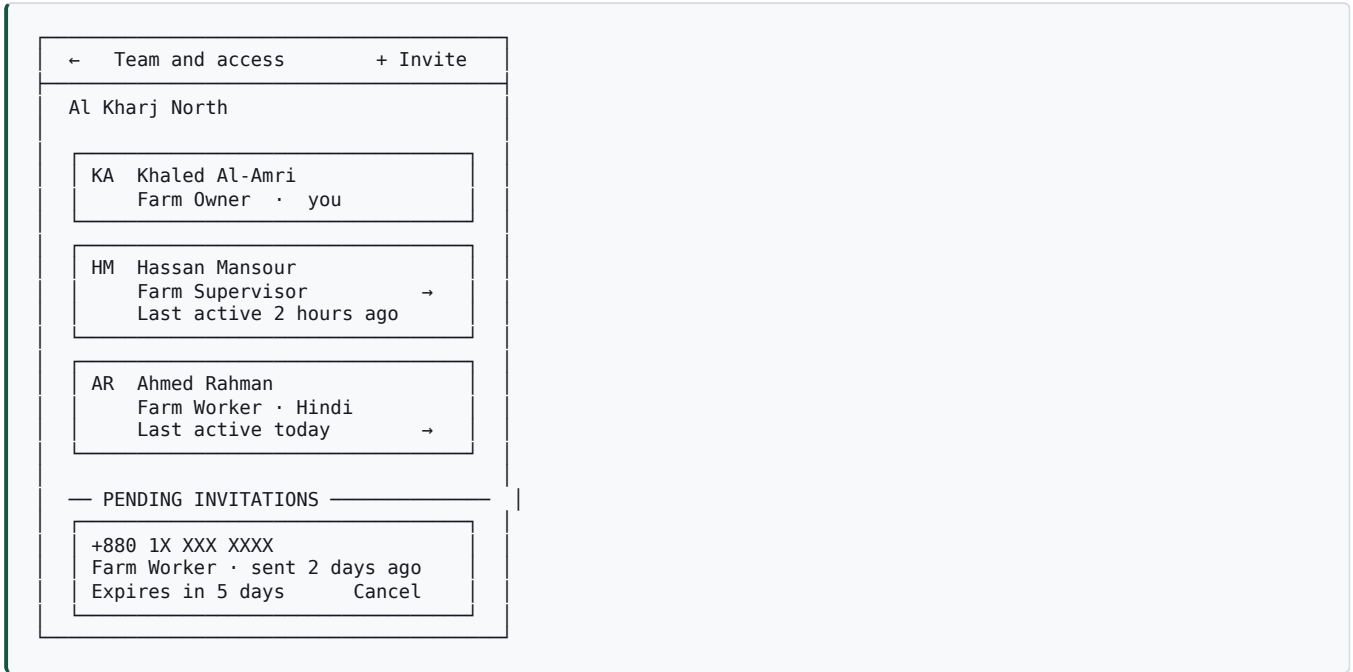
WF5.164 — Reports are shareable through the OS share sheet (WhatsApp, email, file save).

WF5.165 — The **Irrigation: advised vs applied** report is not in the existing report set and must be built. Per plot per period: advised volume, applied volume, difference, and the resulting health trend.

WF5.166 — Reports outside the client's plan appear locked. Do not hide.

WF5.167 — Two summary reports sit alongside the weekly farm report: a **health summary** across plots and trees, and an **irrigation summary** of what was advised, what was applied and what it did to plot health.

5.10.2 F2–F4 — TEAM AND ACCESS



WF5.168 — Pending invitations are listed with their expiry and a one-tap cancel. An owner must always be able to see who has been invited and revoke it.

WF5.169 — Each member row shows their role and their app language, so the assigner knows which language a task will be delivered in.

WF5.170 — Invite (F3) is by mobile number (primary), email, link, 6-character code or QR — see §4.1. Role is selected during the invite and cannot be left blank.

WF5.171 — Invitations may be scoped to specific farms, not necessarily the whole account.

WF5.172 — QR-code invitations produce single-use, short-expiry, role-scoped tokens.

WF5.173 — Removing a member takes effect immediately, invalidates their active sessions for that farm, and is recorded in the activity log. Their historical task completions and observations are **retained and remain attributed** to them.

5.10.3 F5/F6 — SUBSCRIPTION



WF5.174 — An account holds **one subscription** covering every farm on it, up to the 10-farm cap. The screen breaks the price down by what is being charged for — crop area and tree count — so the farmer can see how the total was reached.

WF5.175 — Trial status shows days remaining, prominently, from day one.

WF5.176 — Where the subscription was bought **in the app**, purchase and renewal run through Apple In-App Purchase and Google Play Billing, and billing is managed in the store (§9.1.3).

WF5.177 — Entitlement is **always determined by the server**, never by the local receipt alone, and never by which path the subscription was bought through. The app asks the backend what the user may do. See §9.5.

WF5.178 — Where the subscription was bought **on the web**, this screen states that billing is managed outside the app and offers no purchase or upgrade control. It **does not link to the web purchase page**.

WF5.179 — The *Contact Wafra* route for invoicing, annual contracts and team seats is informational only — it opens the contact screen (F13). It must not describe prices or take payment.

5.10.4 F7–F10 — SETTINGS

WF5.180 — Language, units and region (F8): app language; **land unit** (dunum, hectare, acre); **distance unit** (metres and kilometres, or feet and miles); temperature (always Celsius); date calendar (Gregorian, Hijri or both); and currency display.

WF5.181 — The unit choice made at A7 (§4.8) is the starting value here and can be changed at any time. Changing it re-renders every figure in the app immediately.

WF5.182 — The setting covers the **measurement system generally**, not area alone.

WF5.183 — Notifications (F9): per-channel and per-category toggles, quiet hours, and a channel preference for tasks (push / WhatsApp / SMS). See §7.

WF5.184 — Data and storage (F10): current cache size, "download imagery on Wi-Fi only" toggle, "clear cached imagery" action, and a **pending sync list** showing exactly what has not yet uploaded with a manual *Sync now* button.

WF5.185 — A **shared device** toggle, which shortens the session to 12 hours and requires re-authentication on app launch.

WF5.186 — Settings includes *Delete my account*, which explains what will be deleted and what will be retained, requires confirmation, and creates a deletion request processed within 30 days. This is required by both app stores and by regional data protection law.

5.10.5 F11 — ACTIVITY LOG

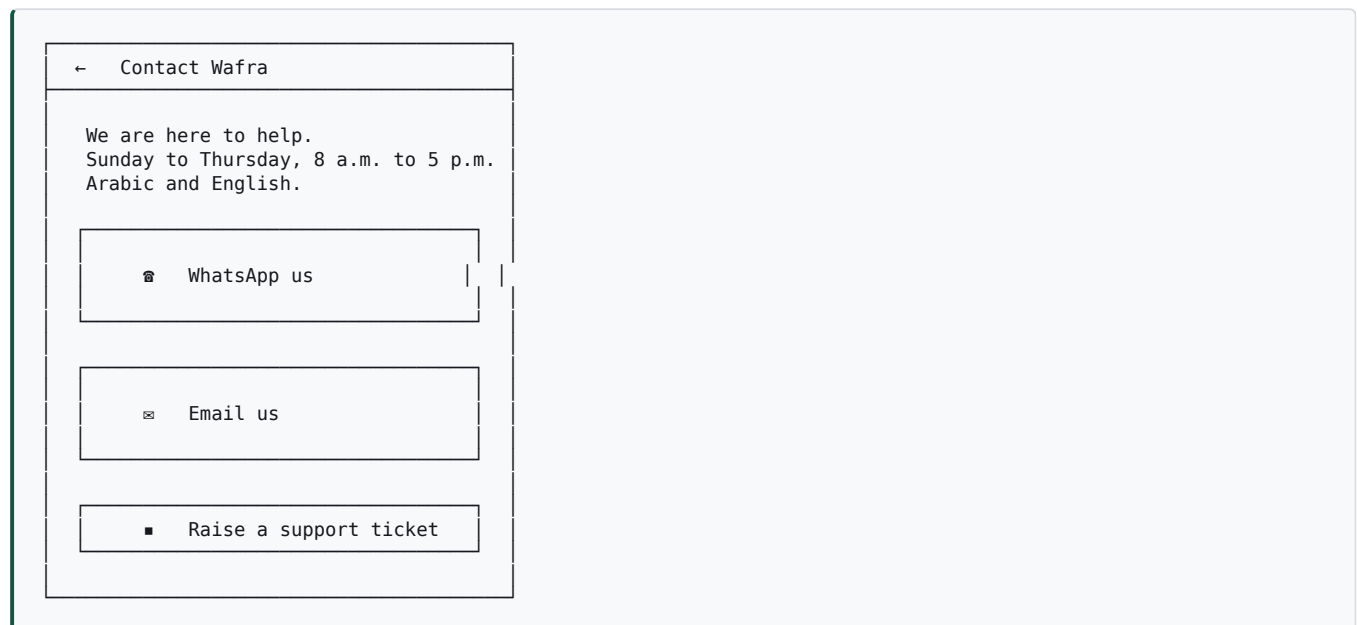
WF5.187 — Owner-only. A chronological, filterable, read-only record of: boundary changes, crop cycle changes, task creation and completion, input logging, member additions and removals, role changes, subscription changes, and advice acknowledgements. Each entry records who, what, when and from where (farm context).

WF5.188 — The activity log is **append-only**. No user, including an Owner, can edit or delete entries.

5.10.6 F12/F13 — HELP AND CONTACT WAFRA

WF5.189 — Help contains a searchable user guide, in all five languages, with short illustrated articles. The guide is served from the backend so it can be updated without an app release.

WF5.190 — Contact Wafra presents exactly two large buttons plus a support ticket option:



WF5.191 — WhatsApp opens a chat with **+966 54 810 0443**, pre-filled with a short message containing the user's account reference, app version and current screen.

WF5.192 — Email opens the device mail client to **info@wafragreen.com**, with the same diagnostic details in the body.

WF5.193 — Both destinations are configuration values fetched from the backend, not compiled into the app, so they can change without an app release.

WF5.194 — The support ticket route uses the existing ticketing capability where the plan includes it. Tickets are routed to Wafra as first line.

6 Advisory layer

Every recommendation the app shows is produced by the advisory layer, which MMC builds and operates. This chapter states what that layer must produce, record and withhold. It does not specify how the models work.

The screens that present advice are in §5.8. This chapter covers what sits behind them.

6.1 What the platform already provides

Derived from MMC's crop feature list. Wafra plan names used.

Crop

Advisory capability	From plan
Crop disease directory	Basic
Operations advisory	Basic
Expert opinion request	Basic
Disease forecasting	Basic
Fertiliser insights	Basic
ET / ET0 / GDD data	Basic
Irrigation scheduler	Pro
Photo-based disease detection	Pro
Growth stage modelling	Pro
Farm anomaly detection	Pro
Custom weather alerts	Pro
Connect with Agro Doctor	Pro
VRA maps (sowing, nitrogen, P&K)	Pro

Trees

Advisory capability	From plan
Tree disease directory	Basic
Seasonal planner	Basic
Satellite-based disease forecasting	Pro
Pest and disease alert	Pro
Early stress warning	Pro
Stress detection	Pro
Ripeness stage distribution	Pro

Advisory capability	From plan
Progressive harvest alert thresholds	Pro
Harvest alert queue	Pro
Tree irrigation schedule	Pro
Irrigation efficiency rating	Pro
Connect with Agro Doctor	Pro

Capabilities that exist only in MMC's Advanced tree tier are not sold and must not be surfaced (WF9.006). The combined **fertigation scheduler** likewise sits above the crop tier behind Wafra Pro and is not offered.

6.2 What must be built

Nothing in the existing feature lists covers the following.

Why it does not exist today	
Immutable advisory log	No equivalent in the feature lists
Product register, per country	The disease directories describe diseases, not registered products, rates or pre-harvest intervals
Advised-versus-applied record	No equivalent
Per-crop attribution on intercropped plots	Not addressed
Worker records and outbound SMS / WhatsApp	No equivalent

6.3 The advisory log

WF6.001 — Every recommendation is written to an advisory log at the moment it is generated, before it is delivered to any user. The record contains: the rendered advice text, the language it was rendered in, the rule version, a snapshot of every input value used, the plot or trees it applies to, the crop it applies to, the recipients, the delivery channels and the timestamp.

WF6.002 — The advisory log is **write-once**. No update or delete path exists at any level, including database administration procedures. It is stored separately from the operational database, with its own backup schedule.

WF6.003 — Retention is **7 years**. Account deletion does not remove advisory log entries; it removes the personal identifiers attached to them.

WF6.004 — A superseded recommendation is marked superseded and linked to the one that replaced it. Neither record is deleted.

WF6.005 — The log is exportable, on request, for a named client and date range, in a documented machine-readable format.

WF6.006 — Acknowledgement and action are recorded against the log entry: who saw the advice, when, and what they recorded doing about it (§5.8.6).

6.4 Registered product register

WF6.007 — Wafra owns the product register — plant protection products, nutrition products and their sources. Where MMC builds the logic that consumes it, MMC sources the register from Wafra and does not maintain a register of its own.

WF6.008 — Wafra maintains the register over time. It is not a one-time handover. The update path is a documented interface Wafra can supply against, and every update takes effect server-side without an app release.

WF6.009 — Each change to the register content is handled as a **change order** to MMC, with its own record. The register is expected to change repeatedly over the life of the product.

WF6.010 — A register of plant protection products, held per country, containing at minimum: active ingredient, registered product name, registration number, registrant, approved crops, approved rate range and unit, pre-harvest interval in days, re-entry interval, and the date the entry was last verified.

WF6.011 — A product is only shown to a user whose farm is in a country where that product is registered. A product not registered in that country is never displayed, in any language, in any screen or report.

WF6.012 — Recommendations are expressed by **element or active ingredient** — N, P, K, calcium and the rest — with the rate. Commercial product names differ country to country and are never the headline. A product name appears only where the register holds a verified registration for that country.

WF6.013 — The pre-harvest interval is taken from the register for the user's country, and the earliest safe harvest date is calculated and presented as a date, not as a number of days (§5.8.4).

WF6.014 — Register entries are versioned with an effective-from date. A recommendation records which version it used, so past advice can be reconstructed against the rules that applied at the time.

WF6.015 — The register is updated server-side and takes effect without an app release.

WF6.016 — Every change to the register is recorded with who changed it, when, and what changed.

WF6.017 — Where an entry has not been verified within 12 months, product names from that entry are withheld and the recommendation falls back to active ingredient and rate only.

6.5 Rule versions and reproducibility

WF6.018 — Every recommendation carries a `rule_version`. A given rule version, given the same inputs, produces the same recommendation.

WF6.019 — Every input used is stored with the recommendation, with its value, unit and source. This is what populates the *Why* section on the advice screens (§5.8) and what makes a past recommendation reconstructable.

WF6.020 — Assumptions a user can correct — irrigation efficiency, soil type, system flow rate — are stored against the plot, not against the recommendation, and a correction triggers recalculation from the next generation onwards. The superseded recommendation is retained (WF6.004).

6.6 When advice is withheld

WF6.021 — Where the client's plan does not include a given advisory capability, no recommendation of that type is generated. The app shows a locked card rather than an empty tab (§5.8).

WF6.022 — Where canopy separation cannot be computed for an intercropped plot on a given date, per-crop recommendations are suppressed for that date and the whole-plot reading is shown marked *Combined canopy* (§5.4.5).

WF6.023 — Where a mandatory input is missing, no recommendation is generated with a substituted value. The app states which input is missing and offers the screen that collects it.

WF6.024 — Crop protection recommendations above a defined risk threshold — rate at the top of the approved range, a pre-harvest interval inside the expected harvest window, or a product flagged as restricted — are held for agronomist review before delivery. The threshold is configuration, not code.

6.7 Liability controls

WF6.025 — Every advisory detail screen carries a non-dismissible line stating that the recommendation is decision support and not a prescription, in the user's language.

WF6.026 — Every crop protection recommendation carries a non-dismissible instruction to check the product label and local regulations before applying.

WF6.027 — The Terms of Use state that recommendations are advisory, that the farmer remains responsible for what is applied, and that legal compliance with local pesticide regulation rests with the applicator. Accepted at registration (§4.7).

WF6.028 — Advice is never presented as certain where it is not. Automated disease identification carries a confidence level and can be rejected by the user; forecasts state the window they cover.

7 Notifications

7.1 Channels

WF7.001 — Three channels: **push notification**, **WhatsApp**, and **SMS**.

WF7.002 — Every notification is sent in the **recipient's** saved language, not the sender's, not the account's. This is a hard dependency and it affects the backend, not just the app.

WF7.003 — WhatsApp delivery uses the WhatsApp Business API with pre-approved message templates. Approval takes time and must be started in week 1. Templates required at launch: task assigned, task overdue, urgent advice, weather alert, weekly report ready, invitation.

WF7.004 — SMS is the fallback where WhatsApp delivery fails or the recipient has not opted in.

WF7.005 — For a **worker record** the channel is not a fallback chain but an explicit per-person choice: SMS on or off, WhatsApp on or off, at least one on (\$5.6).

7.2 Categories and defaults

Category	Default channel	Can user disable?
Task assigned to me	Push + WhatsApp	No (can change channel)
Task overdue	Push	Yes
Urgent advice	Push + WhatsApp	No
Advice needing attention	Push	Yes
Weather alert — severe	Push	No
Weather alert — routine	Push	Yes
Weekly report ready	Push + email	Yes
Farm survey ready	Push + WhatsApp	Yes
Team member joined	Push	Yes
Trial ending	Push + email	Yes
Marketing	Off by default	Yes

WF7.006 — Categories marked "no" above are safety- or contract-critical and may not be switched off entirely, though the user may change channel.

WF7.007 — Quiet hours are user-configurable, default 9 p.m. to 5 a.m. local, and apply to every category except severe weather alerts and urgent advice.

WF7.008 — Every notification carries a deep link that opens the exact object it concerns — task, recommendation, plot, farm, report or invitation — from a cold start, and after login where the user is signed out.

WF7.009 — Notification volume is capped at 6 per user per day, excluding task assignments and severe weather. Beyond the cap, items are batched into a single digest.

8 Roles and permissions

8.1 The model

WF8.001 — Permissions are implemented as a **capability matrix**. Three named role templates ship at launch, each being a named set of capabilities. Adding a fourth role later must not require re-architecting.

WF8.002 — Capabilities are the unit of enforcement, not role names. Application code checks `can(user, 'task.assign', farm)`, never `if (user.role === 'supervisor')`.

8.2 Capability matrix

Capability	Owner	Supervisor	Worker
farm.view	✓	✓	✓ (assigned farms)
farm.create	✓	—	—
farm.edit	✓	✓	—
farm.delete	✓	—	—
farm.transfer	✓	—	—
farm.boundary.edit	✓	✓	—
plot.create	✓	✓	—
plot.delete	✓	—	—
croppcycle.manage	✓	✓	—
analytics.view	✓	✓	—
advice.view	✓	✓	—
advice.acknowledge	✓	✓	—
input.log	✓	✓	✓ (own tasks only)
task.view.all	✓	✓	—
task.view.own	✓	✓	✓
task.create	✓	✓	✓
task.assign	✓	✓	—
task.complete	✓	✓	✓ (own only)
observation.create	✓	✓	✓
tree.view	✓	✓	✓ (assigned)
tree.override	✓	✓	—

Capability	Owner	Supervisor	Worker
report.view	✓	✓	—
report.export	✓	✓	—
worker.manage	✓	✓	—
member.invite	✓	✓ (Worker only)	—
member.remove	✓	—	—
member.role.change	✓	—	—
subscription.view	✓	—	—
subscription.manage	✓	—	—
auditlog.view	✓	—	—

WF8.003 — A Supervisor may invite Workers but not other Supervisors or Owners.

WF8.004 — The structure to model is **owner** → **supervisor** → **a small number of workers**. In Saudi Arabia the supervisor owns the task and reports completion upward; the app must support that without requiring it.

WF8.005 — `worker.manage` covers creating, editing and deactivating **worker records** (§5.6) and is distinct from `member.invite`, which creates app accounts.

WF8.006 — A Worker may create a manual task, but may not assign one to anyone else (§5.8.1).

WF8.007 — Access is granted **per farm**, not per account. A user may be a Supervisor on one farm and have no access to another under the same Owner.

WF8.008 — A farm may have more than one Owner. Ownership transfer and the addition of a second Owner are explicit, SMS-confirmed actions.

8.3 Enforcement

WF8.009 — Every capability is enforced server-side. Every endpoint independently verifies the caller's capability for the specific object being acted on.

WF8.010 — Session tokens carry the user identity only. Role and entitlement are resolved server-side on every request, never trusted from the token payload, so that a role change or a cancelled subscription takes effect immediately rather than at next login.

8.4 Invitations

WF8.011 — Invitation tokens are single use, expire after 7 days, are bound to the role and farm scope chosen at creation, and are revocable at any time by the Owner.

WF8.012 — An invitation link that has been used, expired or revoked shows a clear message and a *Contact the farm owner* action. It never fails silently and it never grants partial access.

9 Plans and entitlements

9.1 The plan structure

WF9.001 — Two levels, basic and pro, in three services. There is no third level.

Service	Levels	Offered when	Priced on
Crops	Basic → Pro	The account has crop farms only	Included crop area
Trees	Basic → Pro	The account has tree farms only	Included tree count
Combined	Basic → Pro	The account has both, or any mixed farm	Both, on one bill

WF9.002 — Plan names in the interface are exactly **Basic and **Pro**.** No plan is called Advanced, Professional or Enterprise anywhere the farmer can see.

WF9.003 — Every feature is either "basic and pro" or "pro only". There is no third state and no feature that appears at one level and disappears at another.

WF9.004 — Wafra's names and MMC's names for the same tier are different and must be reconciled in configuration, never in the app:

Service	Wafra name	MMC tier
Crops	Basic	Professional
Crops	Pro	Advanced
Trees	Basic	Basic
Trees	Pro	Professional

WF9.005 — The mapping above is **server-side configuration.** The app never contains a string from MMC's vocabulary and never derives capability from a tier name.

WF9.006 — Nothing that exists only in MMC's Advanced tree tier is sold. On the tree side that removes: yield estimation, orchard yield optimisation, plot-level harvest planning, personalised AI advisory, canopy structure per tree, hyper-local plot weather, custom weather alerts, higher-resolution satellites and the detailed monthly report. These are out of scope for this release, in the app and in the specification, and the screens that presented them are gone.

WF9.007 — The exclusion is **tier-specific, not feature-name-specific.** Where the same capability sits inside the crop tier behind Wafra Pro — higher-resolution satellites, custom weather alerts, the detailed monthly report — it remains available on crop Pro and appears in §9.3. Scouting (E6) is a Wafra-built screen and is unaffected.

WF9.008 — The app does not estimate yield. Yield figures recorded by the farmer against a crop cycle are history, not prediction, and no forecast is derived from them.

WF9.009 — The subscription sits at **farm level and resolves to one account-level entitlement.** The account holds one subscription whatever the mix of farms.

9.1.1 PRICING

WF9.010 — Price is computed from what the survey detected, never from a static list: crop capability per unit of included area, tree capability per included tree.

WF9.011 — Rates, currencies and any volume banding are server configuration, fetched at launch and on resume. The app performs the arithmetic and shows its working (§4.12) but holds no rate of its own.

WF9.012 — Volume banding on the supplier side is **aggregated across all Wafra clients**, not per end client: Wafra buys as a single customer. No banding is applied per account inside the app, and no farmer sees a volume tier.

WF9.013 — A price is never displayed for a farm whose survey has not completed and been confirmed (§4.10.4).

WF9.014 — Crop area is priced per acre or per hectare and tree capability per tree. The figure shown to the farmer is expressed in **their** chosen unit (§10.4).

9.1.2 COMBINED HOLDINGS

WF9.015 — An account holds **one plan at a time**, whatever mix of farms it contains. Two concurrent subscriptions are never created.

WF9.016 — The service offered is derived from the account's farms, not chosen by the user: crop farms only → Crops; tree farms only → Trees; anything else → Combined.

WF9.017 — Adding a farm of the other type to a single-service account puts that farm in a **pending** state: it is visible, its boundary is kept, and its analytics are replaced by the upgrade path to Combined. Nothing already paid for is affected.

WF9.018 — Archiving every farm of one type offers a downgrade from Combined to the remaining single service, effective at the next renewal.

WF9.019 — Trees are counted for entitlement at the account level across all tree farms, and crop area likewise. A Combined plan resolves to one feature-key set covering both.

9.1.3 HOW THE SUBSCRIPTION IS BOUGHT

Two paths, both compliant with App Store and Google Play policy in every market Wafra sells into. **Redemption codes, licence keys, vouchers and expo codes are not used anywhere** — Apple's Guideline 3.1.1 names them explicitly among the mechanisms an app may not use to unlock content, and the external-purchase-link entitlements that would work around it do not exist in Saudi Arabia or the UAE.

WF9.020 — **Path A — in-app purchase.** A user who registers inside the app buys through StoreKit or Google Play Billing at the standard commission, which is absorbed into the rate. This path must exist for the app to be accepted at all.

WF9.021 — **Path B — web purchase with a server-side entitlement.** A user who is signed in by email buys the licence on the web, outside the app. The entitlement is written against their account and **verified server-side** when they next open the app. There is no code to enter and no unlock step: they sign in and the subscription is simply active.

WF9.022 — Path B depends on the account carrying an email address, which is why email is mandatory on the Owner role (§4.7).

WF9.023 — **The app must not advertise, describe, link to or hint at the web purchase path from inside the app** in Saudi Arabia or the UAE. Path B is discovered outside the app.

WF9.024 — The registration flow leads to Path A. A web purchase produces **no purchase screen in the app at all**.

WF9.025 — The app reads entitlement state at launch and on resume, caches the last known state for offline use, and applies expiry, renewal and cancellation from the server within one refresh cycle.

WF9.026 — Offline, the app honours the last entitlement state it holds and shows its age. It never grants access it has not seen the server confirm, and never revokes access on a failed refresh alone.

WF9.027 — Path C — government and bulk. Volume sales to an organisation buying seats run through **Apple Business Manager / Volume Purchase Program** and **managed Google Play for Work**, under a procurement contract. It is a distinct distribution path with its own entitlement provisioning and is not folded into either consumer flow.

WF9.028 — A seat provisioned under Path C is entitled by the same server-side mechanism as Path B. The app does not know which path paid for it.

9.2 Trial

WF9.029 — Every new account receives a **30-day free trial** on the plan they select at A13, with the full features of that plan. There is no permanently free tier.

WF9.030 — Trial state is held server-side. Reinstalling the app, or signing up again on the same mobile number, does not produce a second trial.

WF9.031 — Trial expiry warnings at 7 days, 3 days and 1 day remaining, by push and email.

WF9.032 — On trial expiry without a subscription, the account enters read-only state: the user can still see their farms, boundaries, history and past reports, but gets no new analytics, no advice and no task creation. Data is retained for 12 months before deletion, and the user is told so explicitly.

9.3 Feature availability — crop plans

Wafra names in bold. Wafra Basic is MMC's Professional tier; Wafra Pro is MMC's Advanced tier.

Capability	Basic	Pro
Satellite resolution	10 m, 3 m+	10 m, 3 m, 1 m
Base satellites (S1, S5)	✓	✓
Higher-resolution satellites (S6)	—	✓
Cloud-free data	—	✓
Add farms	✓	✓
Farm listing and search	✓	✓
Cadastral import	—	✓
Current weather + maps	✓	✓
Weather forecast	14 days	14 days
Historical weather	✓	✓
Custom weather alerts	—	✓
ET / ET0 / GDD data	✓	✓
Advanced GIS layers	—	✓

Capability	Basic	Pro
Farm diary	✓	✓
Dashboard	advanced	advanced
Scouting	✓	✓
1-year progress graph	✓	✓
Weekly farm report	✓	✓
Guaranteed report delivery time	✓	✓
Growth stage modelling	–	✓
Detailed monthly report	–	✓
Anomaly detection	–	✓
Compare last 5 years	–	✓
Soil moisture / temperature	to 1 m	to 3 m
Fertiliser insights	✓	✓
Irrigation scheduler	–	✓
Disease directory	✓	✓
Disease forecasting	✓	✓
Photo-based disease detection	–	✓
Basic maps	✓	✓
Advanced / topographic / compare maps	✓	✓
VRA maps (sowing, N, P&K)	–	✓
Irrigation map	–	✓
Operations advisory	✓	✓
Expert opinion request	✓	✓
Agro Doctor	–	✓
Time series viewer	✓	✓
User guide, support tickets	✓	✓
Multi-user access and roles	✓	✓
Five languages	✓	✓
Tasks	✓	✓
Offline capture	✓	✓
Contact Wafra	✓	✓

9.4 Feature availability — tree plans

Capability	Basic	Pro
Individual tree mapping and crown detection	✓	✓
Tree count per plot	✓	✓
Tree species identification	—	✓
Planting gap detection	—	✓
Plant density mapping	—	✓
Add plots by drawing	✓	✓
Nutrition status, overall health, water stress	✓	✓
Fruit-tree health, photosynthesis efficiency	—	✓
Early stress warning	—	✓
Chlorophyll / nutrient content	—	✓
Water availability, stress detection	—	✓
Base satellites (S1, S5)	✓	✓
Cloud-free (SAR) data	—	✓
Tree disease directory	✓	✓
Satellite disease forecasting	—	✓
Ripeness stage distribution	—	✓
Progressive harvest alert thresholds	—	✓
Average canopy density and vigour	✓	✓
Average chlorophyll, canopy water content	—	✓
Stress status, tree health score, canopy vigour	✓	✓
Chlorophyll, water content per tree	—	✓
Tree list and per-tree card	✓	✓
VRA maps — planting, nitrogen	—	✓
Current weather, 7-day forecast	✓	✓
15-day forecast	—	✓
ET / ET0	—	✓
Tree irrigation schedule	—	✓
Irrigation efficiency rating	—	✓
Seasonal planner	✓	✓
Compare plots over time	—	✓
Agro Doctor	—	✓

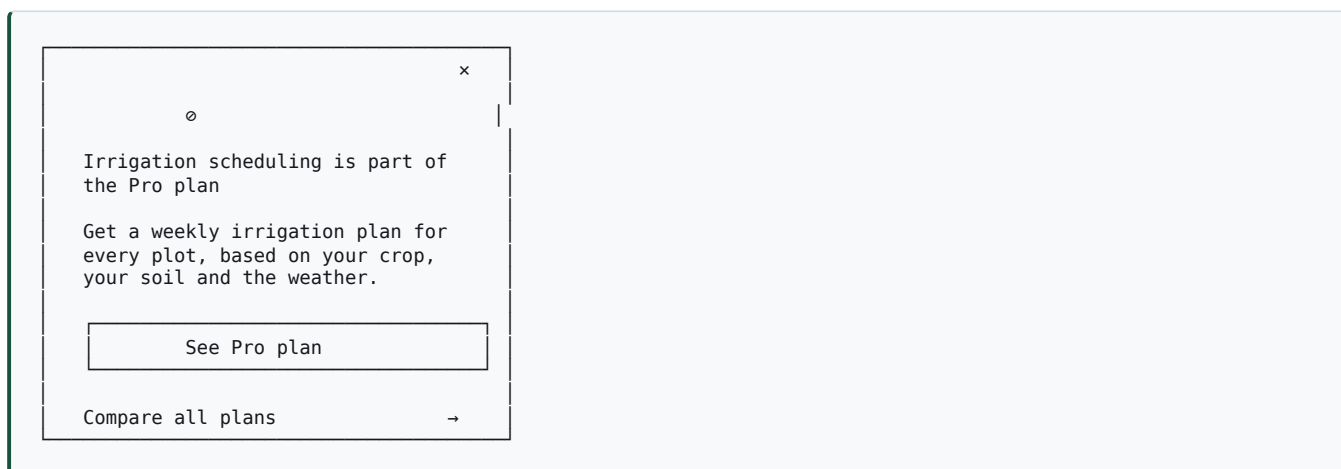
Capability	Basic	Pro
Harvest alert queue	—	✓
Pest and disease alerts	—	✓
Tree and plot dashboards	✓	✓
Health dashboard and alerts	—	✓
Weekly farm report	—	✓
User guide	✓	✓
Support tickets	—	✓
Multi-user, languages, tasks, offline, contact	✓	✓

Wafra Basic is MMC's Basic tree tier; Wafra Pro is MMC's Professional tier. Capabilities that exist only in MMC's Advanced tree tier are not sold and are not specified (WF9.006).

9.5 How locked features behave

WF9.033 — A feature outside the user's plan is **visible and locked**, never hidden, unless it would be meaningless (for example, tree screens for a crop-only client).

WF9.034 — A locked feature shows a lock icon in place of its value or content. Tapping it opens a single, consistent upgrade sheet:



WF9.035 — The upgrade sheet names the plan required and describes the benefit in one sentence of plain language. It never shows a technical feature name.

WF9.036 — Entitlement checks happen server-side on every request. A modified client must not be able to obtain data outside its plan.

10 Internationalisation (i18n)

10.1 Languages

WF10.001 — Five languages ship at launch: **Arabic, English, Hindi, Bengali, Pashto**. Arabic is listed first everywhere the five appear.

WF10.002 — In the language picker each language appears **in its own script only** — no English gloss, no second line (§4.3).

WF10.003 — **Arabic and Pashto are both right-to-left**. Pashto is written in a modified Perso-Arabic script and requires full RTL treatment.

WF10.004 — RTL treatment includes: layout mirroring, navigation and back-arrow direction, list chevrons, progress and slider direction, chart axis direction, icon mirroring where an icon implies direction, and correct bidirectional handling of mixed Latin/Arabic strings such as "P-04" inside an Arabic sentence.

WF10.005 — Numbers use **Western Arabic numerals (0–9)** in all languages including Arabic, unless the user chooses Eastern Arabic numerals (٠-٩) in Settings. Western numerals are the norm in commercial agriculture across the region.

WF10.006 — Fonts covering Devanagari (Hindi), Bengali and Perso-Arabic (Pashto) are bundled with the application, not relied upon from the device.

WF10.007 — Test on a physical Android 8 device in each script before sign-off.

WF10.008 — Known rendering faults to fix and to test for explicitly: an English label repeated beneath an English label, Arabic text misaligned inside a mirrored layout, and any debug marker such as a visible "RTL" string reaching a build.

WF10.009 — Language switching takes effect immediately across the entire interface, without restarting the app.

WF10.010 — Layouts accommodate a 40% expansion in string length without breaking. Emulator rendering is not evidence.

10.2 String catalogue

WF10.011 — All strings live in an i18n platform with stable keys. No English text in application source. This includes push notification bodies, SMS text, WhatsApp template variables, report headings, PDF labels, error messages and unit abbreviations.

WF10.012 — An **agronomic glossary** is maintained alongside the string catalogue, fixing the agreed translation of terms such as evapotranspiration, crop coefficient, pre-harvest interval, fertigation, canopy, vigour, water stress. These terms must be translated identically everywhere, and by someone who knows what they mean.

WF10.013 — A named reviewer per language is recorded and signs off the catalogue before release.

WF10.014 — Server-generated content — recommendation text, alert bodies, report sections — comes from the same catalogue as the app, keyed the same way.

10.3 Content in the user's language

WF10.015 — Every notification, task instruction, report and recommendation is rendered in the **recipient's** language, resolved per user at send time.

WF10.016 — Where a translation is missing for a key, fall back to English and log the missing key server-side. Never show a raw key to a user.

10.4 Locale conventions

WF10.017 — Temperature in Celsius. **Water volume in cubic metres**, with litres for per-tree figures. Depth in millimetres is a model input and is not shown as a headline figure (§5.8.2).

WF10.018 — **Times are shown on a 12-hour clock with a.m. and p.m.** The 24-hour clock is not used in the interface, in notifications, or in reports.

WF10.019 — The **land unit is asked at sign-up** and defaults from the country detected on the phone — dunum in the UAE, hectares in Saudi Arabia, acres elsewhere. It is shown with hectares in brackets: 12.4 dunum (1.24 ha). Changing it in Settings swaps the order everywhere. 1 dunum = 0.1 ha throughout.

WF10.020 — Dates display in **both Gregorian and Hijri** when the app language is Arabic, Gregorian primary: 3 Aug 2026 (19 Safar 1448) . The user may reverse the order or show one only. Hijri dates are display-only; all storage and calculation is Gregorian.

WF10.021 — Prices display in the user's local currency — the GCC currencies and the Jordanian dinar — with the USD figure alongside: SAR 375 / month (USD 100) . Exchange rates come from the server and are refreshed daily. The app never calculates a conversion itself.

WF10.022 — Time zones: all timestamps stored in UTC, displayed in the farm's local time zone, which is a property of the farm, not the device. A supervisor travelling must see farm-local times.

11 Offline behaviour

11.1 Principle

WF11.001 — The app is read-anywhere, write-in-the-field. Everything the user has already seen remains viewable offline. A defined set of field actions can be created offline but configuration and structural changes cannot be made offline.

11.2 What is cached

Data	Cached	Retention
Farm boundary, plot records and geometry	✓	Until changed
Current crop cycles	✓	Until changed
Latest measure values per plot	✓	30 days
Latest measure raster for each plot, current measure only	✓	14 days
Map tiles covering the user's crop farms, zoom 12–17	✓	30 days
Map tiles covering the user's tree farms, zoom 12–19	✓	30 days
Tree records and latest health scores	✓	30 days
Tasks assigned to this user, plus all open tasks on their farms	✓	Until closed + 7 days
Active recommendations	✓	Until superseded
Weather: current + forecast at last sync	✓	3 days
The last 4 reports	✓	90 days
User guide	✓	Until updated
String catalogue for the user's language	✓	Until updated
Historical imagery beyond the latest	✗	—
Other users' tasks	✗	—

WF11.002 — Total cache is capped at **500 MB** by default, user-adjustable between 100 MB and 2 GB in Settings. When the cap is reached, evict oldest imagery first, never queued user data.

WF11.003 — Queued user data (photos, completions, observations) is **never evicted** and does not count against the cache cap.

11.3 What can be created offline

Action	Offline
Complete a task, with photo, quantity and note	✓

Action	Offline
Create a field observation with photos	✓
Record an input applied (D7)	✓
Read any cached screen	✓
Create or assign a task	✗
Confirm a survey result (A11)	✗
Edit a plot boundary	✗
Add or edit a crop cycle	✗
Invite a member or change a role	✗
Change subscription	✗

WF11.004 — Actions that cannot be performed offline show a clear, calm message naming what is needed ("You need a connection to assign a task"), not a disabled control with no explanation.

11.4 Sync and conflict

WF11.005 — Every offline-created record is written locally with a client-generated UUID and an idempotency key. Re-sending the same record must never create a duplicate.

WF11.006 — Sync is automatic on regaining connectivity, on app foreground, and on a manual *Sync now*. It is incremental and resumable — a sync interrupted at photo 7 of 20 resumes at photo 7.

WF11.007 — Conflict rules, per object type:

Object	Rule
Task completion	First write wins. If the task was already completed by someone else, the second completion is stored as an additional record and both are visible. Never silently discard a worker's evidence.
Field observation	Always accepted. Observations are additive; there is no conflict.
Input log	Always accepted, additive.
Task assignment or status changed server-side while offline	Server wins. The user is shown a non-blocking notice: "This task was changed while you were offline."

WF11.008 — A sync failure is surfaced to the user in the connectivity indicator and in Settings → Data and storage, with the count of pending items and a retry. It is never silent.

WF11.009 — Photos queued for upload survive the app being force-closed, the device being restarted and the app being updated. They are written to durable storage with their metadata before the user sees a confirmation.

WF11.010 — Upload retries use exponential backoff, resume partial uploads, and prefer Wi-Fi where the user has set that preference — but a task completion older than 24 hours uploads on any connection regardless of the setting.

WF11.011 — The offline store, including cached imagery and queued photos, is **encrypted at rest** on the device.

11.5 What the user sees

WF11.012 — A connectivity indicator is present on every screen, with three visual states: online, offline, and syncing with a count of pending items.

WF11.013 — Cached content older than 24 hours carries a visible age label.

12 Open items

Everything else in this document is settled. These nine are not, and the sections that depend on them cannot be finished until they are answered.

#	Item	Needed from
1	Surveys that never convert. Every prospective customer must trigger a survey to see a price (§4.10.4). Does MMC charge Wafra for a survey that does not become a subscription? At a 10% conversion rate this is a material cost line	MMC
2	Per-field pricing. A farmer who wants one field monitored rather than the whole farm needs a defined menu and a price. Neither exists	Commercial
3	Combined holdings. A client with both crops and trees — one plan at a reduced rate, or two plans bought separately? A13 and F5 assume one (§4.12.1)	Commercial
4	Satellite cadence. The source material gives hourly, every five minutes, and a weekly model run. Imagery refresh and model run frequency are two different things and the app states both to the farmer. Confirm each	MMC
5	Crop nomenclature. "Field crops" with no detail, or a level below it? The current examples sit at inconsistent levels of granularity (§4.9)	Agronomy
6	Worker titles. Does a worker record still need a title or role? The original intent was that the title changes what the app shows — but a worker who never opens the app may not need one (§5.6)	Product
7	Tree positioning. Grid coordinates ("position 6, position 8") read as office-based. Field survey works along rows, linearly, and plantation spacing is inconsistent — some countable, some dense and askew. The positioning model needs rethinking for field use (§5.7.1)	Agronomy
8	Terms of use. Where exactly does acceptance sit? There is an awkward gap between login and the terms agreement (§4.7)	Product
9	Advice headline density. Advisory headlines may still carry too much detail for the field (§5.8)	Product

13 Assumptions

Stated so they can be corrected rather than discovered. If any is wrong because the capability already exists, say so in the feasibility response and the estimate should fall accordingly.

1. The existing crop and tree analytics are available through an interface the app can consume.
2. Task management, the workforce records, the advised-vs-applied record and the report at WF5.165 are net-new development.
3. The farm → farm boundary → plot → tree hierarchy with multiple crops per farm is a change to the current data model, which supports one crop per plot.
4. The land-use survey returns plots and tree counts the app can edit — split, join, redraw, remove and add — before they are committed (§4.10.3).
5. Farm sharing and role-based access is a change to the current product, which is single-user.
6. Five-language support with full RTL is substantially new.
7. Offline capture as specified in §11 is new.
8. Server-side entitlement, resolvable from a purchase made outside the app, is new (§9.1.3).
9. Nothing in this document requires changes to satellite acquisition or to the AI models.

Annex A — Suggested practice

Nothing in this annex is a requirement. It is a working starting point for the areas the specification deliberately does not mandate: how to run the repository, cut a release, submit to the stores, accept the build, and handle hosting and personal data. Where MMC already has practice that works, keep it.

A.1 Repository and branching

- Three long-lived branches: `develop` → `staging` → `main`. No commits directly to `staging` or `main`; everything arrives by pull request with at least one approving review.
- Every pull request runs, and passes: linting, unit tests, a build of both mobile platforms, and dependency scanning. A red build is not merged.
- Semantic versioning `MAJOR.MINOR.PATCH`, with a monotonically increasing build number. The version and build number are displayed in the app (WF5.161) and sent on every API request.

A.2 Configuration

- No environment-specific value compiled into the application except the API base URL, which is set per build flavour. Everything else — contact details, feature flags, minimum version, prices, support hours — fetched from the backend at launch.
- Three build flavours: `dev`, `staging`, `production`, installable side by side on the same device with distinct bundle identifiers, app names and icons.
- Signing keys and store credentials in the CI secret store, backed up in two separate locations. Losing the Android upload key makes the app permanently un-updatable.

A.3 Release checklist

A list worth running for every release, small changes included.

Before the release branch is cut

1. All requirements in the release scope marked done and their acceptance criteria met.
2. String catalogue complete: zero missing keys in any of the five languages.
3. Named reviewer sign-off recorded for each language (WF10.013).
4. RTL layout verified for Arabic and Pashto on a physical device.
5. Bengali, Hindi and Pashto rendering verified on a physical Android 8 device.
6. Offline scenarios tested: airplane mode capture, force-close during upload, sync after 24 hours offline, duplicate-submission test.
7. Entitlement matrix tested for every level of every service, including locked-feature behaviour.
8. Role matrix tested for all three roles, including the server-side rejection of a client that tries to exceed its role.
9. Dependency scan clean of critical findings.
10. Crash-free session rate on staging above 99.5% over the last 7 days.

Cutting the release

11. Merge `develop` → `staging`, tag `vX.Y.Z-rc.N`.

12. Deploy backend to staging. Run database migrations. Confirm migrations are reversible.
13. Distribute the mobile build to TestFlight (internal) and Google Play internal testing.
14. UAT against the Jordanian farms (§A.8). Wafra sign-off.

Going live

15. Merge `staging` → `main`, tag `vX.Y.Z`.
16. Deploy backend to production during an announced maintenance window, outside 05:00–19:00 Arabian Standard Time.
17. Run migrations. Verify with a smoke test of the 10 critical paths (§A.5) before releasing the mobile build.
18. Submit to App Store Review and Google Play. Expect 24–72 hours; see §A.4.
19. Staged rollout on Google Play: 10% → 50% → 100% over 72 hours, watching crash rate.
20. iOS phased release enabled.
21. Publish release notes in all five languages.
22. Tell Wafra support what changed, before users find out.

A.4 App store submission

- Both store listings prepared in **all five languages**: title, subtitle, description, keywords, screenshots and the privacy questionnaire.
- Screenshots showing the app in the language of that listing.
- The privacy questionnaire (Apple Nutrition Label / Google Data Safety) completed accurately for location, photographs, contact details, identifiers and usage data.
- A demo account for review, populated with farms, imagery, advice and tasks. An empty app is rejected as non-functional.
- Subscriptions configured in App Store Connect and Play Console with the 30-day trial, localised prices per territory (SAR, AED, JOD, USD), and clear descriptions. Both stores require trial terms to appear in the app before purchase.
- No link to an external purchase page for the subscriptions sold in-app (WF5.178) — both stores reject it.
- Location permission strings that explain the actual purpose in the user's language, naming both uses: showing position on the farm map, and recording where field photographs were taken.

A.5 Smoke test — the ten paths

Worth running against production after every deployment, on a real device, before announcing the release:

1. Sign up with a new mobile number and email, receive and enter the code.
2. Draw a farm boundary, send it for survey, edit a returned plot, confirm, and see a price.
3. Open a plot; a measure map renders with a date and a legend.
4. Open Advice; a recommendation is listed with its reasoning.
5. Assign an advisory item to a worker record and confirm the message goes out.
6. As that second user, receive the notification in a different language and open the task.
7. In airplane mode, complete the task with a photograph. Confirmation appears immediately.
8. Restore connectivity; the completion and the photograph sync; the supervisor sees both.
9. Generate a weekly report; it downloads as a PDF in the requested language with Wafra branding.
10. Attempt a locked feature; the upgrade sheet appears and the API returns `402`.

A failure in any of the ten is better treated as a rollback than a hotfix-in-place.

A.6 Rollback

- Every backend deployment rollback-capable within 15 minutes. Database migrations written so the previous application version continues to work against the new schema — additive first, destructive only in a later release.
- Every material change behind a server-side feature flag that can be turned off without an app update.
- A forced-update mechanism driven by `minimum_supported_version` in `/config`, showing a blocking screen with a store link. Used only for security fixes and breaking API changes.

A.7 Definition of done

A requirement can reasonably be called done when all of the following are true:

1. It behaves as specified on both platforms.
2. It behaves correctly in all five languages, including RTL.
3. It has an empty state, a loading state and an error state (WF2.011, WF2.012).
4. Its offline behaviour matches §11 — including "cannot be done offline", where that is the specification.
5. Its entitlement and capability behaviour is enforced server-side and verified by a test that calls the API directly, not through the app.
6. It works on a 360 × 640 dp screen at 200% font scale.
7. It has been used, once, by a person who did not build it.

A.8 User acceptance testing

UAT on staging against the **Jordanian farms already monitored by Wafra**, using real boundaries and real imagery history. Two named sign-offs before production release:

Area	Sign-off
Agronomic correctness — advice, units, crop parameters, terminology in Arabic	Hani AlHabbab
Functional correctness — everything in this document	Romain Bigaré

A.9 Data residency

Client data resides in a secure cloud **in the country it originated from** — Emirati farm data in the UAE, Saudi farm data in Saudi Arabia. This is a requirement, not a preference, and it is stated as a requirement in §2.4.

Practical consequences worth planning for: a deployment per country rather than a single regional one; a routing rule that resolves a farm to its jurisdiction at write time, not at read time; and a backup and disaster-recovery footprint that stays inside the same border.

A.10 Backups

- Daily full backup, point-in-time recovery to any moment in the last 7 days.
- Export of all data belonging to a Wafra client, in a documented, machine-readable format, on request, within 10 working days.

A.11 Media and personal data

- Photographs served through short-lived signed URLs rather than public object storage links.
- Every task photograph is tagged with a named worker, a timestamp and a GPS coordinate, which constitutes location processing of identified individuals. That carries consequences the build should plan for:
 - The purpose stated in the privacy policy in all five languages.
 - Workers told, at first launch, in plain language, that photographs they take record their location and are visible to their supervisor and the farm owner.
 - Location recorded only at the moment a photograph or observation is captured, with no continuous tracking of worker position.
 - A route for a worker to request deletion of their personal data.
- Personal data not used for any purpose beyond operating the service without a separate consent.
- Log records containing personal data excluded from application logs and error-reporting tools. Mobile numbers, names and coordinates kept out of crash reports.

Annex B — Requirement register

457 requirements. Section numbers refer to this document.

ID	§	Requirement
WF2.001	2.3	Mobile application for iOS 15+ and Android 8.0 (API 26)+
WF2.002	2.3	Usable on a 5-inch screen at 360 × 640 dp
WF2.003	2.3	Install size below 80 MB
WF2.004	2.3	Minimum touch target 48 × 48 dp, with at least 8 dp between adjacent targets
WF2.005	2.3	The primary action on every screen sits in the bottom third of the display
WF2.006	2.3	Body text no smaller than 16 sp
WF2.007	2.3	Respect the operating-system font-size setting up to 200% without clipping text or losing...
WF2.008	2.3	Status is not to be communicated by colour alone
WF2.009	2.3	Health and status use the scale below, with the same colour, symbol and word every time
WF2.010	2.3	No screen presents more than one primary action
WF2.011	2.3	Every screen that can potentially have nothing to show (e.g. no farms yet, no plots on a...
WF2.012	2.3	Every screen has an explicit loading state and an explicit error state with a plain-language...
WF2.013	2.3	Numbers are always shown with their unit, rounded to actionable precision: "412 m³" instead...
WF2.014	2.3	Icons carry a text label beneath or beside them
WF2.015	2.3	Never block the user with a full-screen "no connection" message when cached data exists
WF2.016	2.3	Every data-entry action that can be performed offline shows immediate local confirmation
WF2.017	2.3	The application carries Wafra branding only : logo, colours, app icon, splash, store listing...
WF2.018	2.4	Client data resides in a secure cloud in the country it originated from
WF2.019	2.4	Jurisdiction is resolved from the farm, not from the user
WF2.020	2.4	Backups and disaster-recovery replicas are held under the same residency terms as primary...
WF2.021	2.4	Where a new market is opened, residency for that market is settled before the first farm is...
WF3.001	3.1	Owner and Supervisor see five tabs:
WF3.002	3.1	Worker sees three tabs, and lands on My Work at launch:
WF3.003	3.1	A badge on the Advice tab shows the count of unread recommendations at Needs attention or...
WF3.004	3.1	A badge on the Tasks / My Work tab shows the count of tasks assigned to this user and due...
WF3.005	3.1	Tab bar labels are always visible
WF4.001	4.2	Registration creates an account with no role
WF4.002	4.2	A user who creates a farm becomes a Farm Owner of it
WF4.003	4.2	A user who accepts an invitation receives the role carried in that invitation — Farm...
WF4.004	4.2	The same person may do both
WF4.005	4.2	A Worker can never create, delete or edit a farm, a plot or a boundary, on any farm, however...
WF4.006	4.2	An invitation is single use, expires after 7 days, and is bound to the role and the farm...
WF4.007	4.2	Four delivery methods, all producing the same single-use token: WhatsApp deep link...
WF4.008	4.2	Inviting someone into the app is a different thing from creating a worker record for someone...
WF4.009	4.2	An Owner sees all pending invitations for their farms with expiry and a one-tap revoke
WF4.010	4.2	Where a Worker has no smartphone of their own, a Supervisor can complete the invitation on...
WF4.011	4.3	Each language is shown in its own script only
WF4.012	4.3	Arabic is first in the list
WF4.013	4.3	All five options fit on one screen without scrolling on a 360 × 640 dp display

ID	§	Requirement
WF4.014	4.3	Pre-selected to the device's OS language if it is one of the five; otherwise Arabic
WF4.015	4.3	Selecting a right-to-left language mirrors the layout immediately, on this screen, before...
WF4.016	4.3	Never shown again after first run
WF4.017	4.4	A2 carries exactly three actions and no input of any kind
WF4.018	4.4	The three routes: Log in → A3, Create an account → A4, Join a farm → A15
WF4.019	4.4	Log in is first, because the operating system usually remembers the credentials and the...
WF4.020	4.4	A language control stays on this screen, so a wrong choice at A1 costs one tap
WF4.021	4.4	Opening an invitation deep link goes straight to A15 and never shows A2
WF4.022	4.5	Mobile number and email are interchangeable identifiers
WF4.023	4.5	Login accepts identifier + password, or identifier + one-time code
WF4.024	4.5	Sessions persist for 180 days of activity on a personal device
WF4.025	4.5	Log out is always available and clears the local cache including cached imagery and queued...
WF4.026	4.6	The tour is a fixed sequence of still images with a caption
WF4.027	4.6	The first snapshot is a satellite view of a typical farm with trees and crops identified on...
WF4.028	4.6	The following snapshots are advisory examples, numbered 1 of 5, 2 of 5 and so on, so the...
WF4.029	4.6	Skip is present on every snapshot and goes straight to A5
WF4.030	4.6	The tour is shown once, on the registration path only
WF4.031	4.6	The tour runs in the language chosen at A1, from the same string catalogue as the rest of...
WF4.032	4.7	Registration takes both a mobile number and an email address for an account that will own a farm
WF4.033	4.7	Email is mandatory on the Owner role
WF4.034	4.7	Verification (A6) is a code, by SMS to the number or by email to the address, verified...
WF4.035	4.7	Country selector defaults from the SIM's country, then device locale, then Saudi Arabia
WF4.036	4.7	The number field accepts spaces and dashes and normalises them
WF4.037	4.7	Terms and Privacy links open in-app, in the user's selected language
WF4.038	4.7	Six-digit code, valid for 10 minutes
WF4.039	4.7	Resend available after 45 seconds
WF4.040	4.7	After 5 incorrect code entries the account is locked for 15 minutes and the user is shown...
WF4.041	4.8	Name is mandatory
WF4.042	4.8	A show/hide control on the password field
WF4.043	4.8	The land unit is asked here, pre-selected from the country detected on the phone: dunum in...
WF4.044	4.8	Password is optional for Worker accounts created by invitation; those may sign in by code alone
WF4.045	4.8	A user who reaches A7 through Create an account becomes the Owner of the first farm they create
WF4.046	4.8	Around 80% of accounts are farm owners and 20% supervisors
WF4.047	4.9	Three options : Field crops, Date palms and fruit trees, Both
WF4.048	4.9	The tree category is called date palms and fruit trees throughout the app. "Orchard" is not...
WF4.049	4.9	The examples under each option are all at the same level — individual crops
WF4.050	4.9	Both carries no example line
WF4.051	4.10	This screen is reachable only by Owners and Supervisors with the plot.create capability
WF4.052	4.10	Both routes are offered with equal weight
WF4.053	4.10	The choice is per farm, not per account
WF4.054	4.10	The survey route is offered again from Farm settings (B11) for a farm that was created by...
WF4.055	4.10	The app opens the satellite map centred on the device's location and lets the farmer scroll...
WF4.056	4.10	A search bar is present and visible at all times on every map screen
WF4.057	4.10	Search is served by the Google Maps API
WF4.058	4.10.1	The manual route supports many plots per farm

ID	§	Requirement
WF4.059	4.10.1	Each hand-drawn plot carries its own crop entry
WF4.060	4.10.1	There is no limit on the number of points in a boundary
WF4.061	4.10.1	Where the farm holds trees, the survey still runs on the drawn plots, because the tree count...
WF4.062	4.10.1	Boundary auto-detection from a single tap in the middle of a field is not offered
WF4.063	4.10.1	Opens centred on the device's GPS position at approximately 1:5,000
WF4.064	4.10.1	Satellite basemap by default
WF4.065	4.10.1	Live area readout in the user's chosen unit, with hectares in brackets
WF4.066	4.10.1	Undo removes the last placed vertex
WF4.067	4.10.1	Vertices are draggable with a touch target of at least 48 dp, offset above the fingertip so...
WF4.068	4.10.1	Self-intersecting polygons are rejected with a plain-language message and the offending...
WF4.069	4.10.1	A polygon smaller than 0.1 ha or larger than 10,000 ha triggers a confirmation
WF4.070	4.10.2	The drawing interaction is the one at A9 : same vertex handling, same undo, same area...
WF4.071	4.10.2	One polygon per survey
WF4.072	4.10.2	On Send for survey the farm record is created immediately in a surveying state and the...
WF4.073	4.10.2	The expected wait is stated on the button and on the farm card: 15 to 20 minutes
WF4.074	4.10.2	A survey can be requested without an active subscription and without a payment method on file
WF4.075	4.10.2	Only one survey may be in flight for a given farm
WF4.076	4.10.2	A survey that has not returned within twice the stated time shows a plain notice on the farm...
WF4.077	4.10.2	A survey that fails offers two actions : try again, or draw the plots by hand (A9)
WF4.078	4.10.3	Three classes are returned: open field crops including fallow land, date palms and fruit...
WF4.079	4.10.3	Colour on the map is never the only signal
WF4.080	4.10.3	Covered agriculture and structures default to excluded
WF4.081	4.10.3	The farmer can split a plot the survey merged, join plots it separated, edit any boundary,...
WF4.082	4.10.3	Boundary editing here uses the same interaction as A9 and the boundary editor (C5) : tap to...
WF4.083	4.10.3	Every area can be included or excluded, and its class corrected — field crops to trees,...
WF4.084	4.10.3	Totals update as the farmer changes anything: included area, crop area, tree count, excluded...
WF4.085	4.10.3	On Confirm and continue, every included area becomes a plot, carrying its class as the plot type
WF4.086	4.10.3	The classification the algorithm returned and the farmer's corrections are both retained
WF4.087	4.10.3	A survey that returns no farmed area at all shows a plain explanation and two actions : draw...
WF4.088	4.10.3	The farm stays in the surveying state until this screen is confirmed
WF4.089	4.10.4	After A11 the farmer goes to A12 (farm details) and then to A13, where the service and the...
WF4.090	4.10.4	The price is calculated from what the survey found
WF4.091	4.10.4	No price is shown before the survey has completed and been confirmed
WF4.092	4.10.4	Where the farmer changes the included areas at A11, the price at A13 recalculates from the...
WF4.093	4.10.4	A13 is pre-selected from what the survey found — crops only, trees only or both — and the...
WF4.094	4.10.4	Changing the service at A13 does not discard the excluded or unselected areas
WF4.095	4.11	Only farm name and farm type are mandatory
WF4.096	4.11	The irrigation system is not collected
WF4.097	4.11	Soil type may be left blank; the backend fills it from the soil layer where available and...
WF4.098	4.12	The plan chooser shows only the service matching the farm just created, and the price it...
WF4.099	4.12	The plan cards carry a live calculation, not a static price: the quantity the survey found,...
WF4.100	4.12	Both a monthly and an annual figure are available on each card
WF4.101	4.12	Two levels per service, always in the order Basic → Pro
WF4.102	4.12	Rates are fetched from the server, never hard-coded, and displayed in the currency of the...
WF4.103	4.12	Compare all features opens the full comparison table (F6)

ID	§	Requirement
WF4.104	4.12	Adding a sixth farm shows an enterprise screen inviting the farmer to talk to an advisor,...
WF4.105	4.12	An account holds a maximum of 10 farms
WF4.106	4.12.1	Where the account has at least one crop farm and at least one tree farm, the plan chooser...
WF4.107	4.12.1	A combined plan is a single purchasable product with a single price and a single renewal date
WF4.108	4.12.1	A mixed farm — crops and trees inside the same farm boundary — requires a combined plan
WF4.109	4.12.1	Where a client on a crop-only plan adds their first tree farm, the app offers an upgrade to...
WF4.110	4.12.1	Entitlement is resolved server-side as a single set of feature keys
WF4.111	4.12.1	Downgrading from a combined plan to a single service is only possible once the farms of the...
WF4.112	4.13	A confirmation screen stating when the first satellite imagery will be available, and a...
WF4.113	4.14	Opening an invitation deep link, with or without the app installed, leads to: install if...
WF4.114	4.14	An invited user never sees A8–A13
WF4.115	4.14	Invitations expire after 7 days and are single use
WF4.116	4.14	A used, expired or revoked invitation shows a clear message and a Contact the farm owner action
WF4.117	4.14	On successful join, a Worker lands directly on My Work; a Supervisor lands on Home filtered...
WF5.001	5.1	The summary card at the top counts farms by worst-plot status, not by average
WF5.002	5.1	See what to do navigates to the Advice inbox (D1) pre-filtered to Needs attention and Urgent
WF5.003	5.1	Each farm card shows: status icon and colour, farm name, farm type icon, area, plot or tree...
WF5.004	5.1	"Updated X ago" refers to the age of the most recent satellite imagery, not the last app refresh
WF5.005	5.1	A farm awaiting its survey result shows in place of a status card: "Survey in progress — we..."
WF5.006	5.1	A farm in either survey state is not counted in the summary card at the top and carries no...
WF5.007	5.1	All farms is a first-class view
WF5.008	5.1	The map toggle icon that appears beside each individual farm also appears beside the ALL...
WF5.009	5.1	Farms are sorted by severity first, then by most recently viewed
WF5.010	5.1	Pull to refresh, with a visible refresh button as the non-gesture equivalent
WF5.011	5.1	Empty state: "You have no farms yet" with an Add your first farm button leading to A9
WF5.012	5.2	The health block lists only measures included in the farm's current plan
WF5.013	5.2	Tapping any row in the health block opens the measure viewer (B7) for that measure across...
WF5.014	5.2	The Trees row is present only for tree and mixed farms
WF5.015	5.2	The weather strip shows current conditions and a 7-day or 14-day forecast according to plan...
WF5.016	5.2	Farm settings (⚙, top right) is visible to Owner and Supervisor only
WF5.017	5.2	Workforce sits on the farm screen, in the Explore list beside Plots and Trees, showing the...
WF5.018	5.3	Plots are listed flat under the farm
WF5.019	5.3	The list has a farm selector at the top with All farms as one of its options, so plots can...
WF5.020	5.3	Sort options: Needs attention first (default), Name, Area, Most recently changed
WF5.021	5.3	Each plot row shows the measure value and its change over the last 7 days, with a direction...
WF5.022	5.4	The measure selector is a dropdown listing every measure available on this farm's plan, in...
WF5.023	5.4	Date stepping with ◀▶ moves between available imagery dates, not calendar days
WF5.024	5.4	The interpretation line below the map names where and how long, not only what
WF5.025	5.4	See what to do is the primary action and links to the relevant recommendation (D2–D4, D6)
WF5.026	5.4	The ⋮ menu contains: Edit plot, Edit boundary, Add crop cycle, Add observation, Share plot...
WF5.027	5.4	Plot detail aggregates the whole plot: area, tree count and a severity per measure
WF5.028	5.4	Years are rendered as plain four-digit numbers with no thousands separator. "1990", never...
WF5.029	5.4.1	Full-screen measure map, pinch-zoom and pan, with a persistent legend showing the value...
WF5.030	5.4.1	Tapping any point on the map shows the value at that point in a small callout, with its...
WF5.031	5.4.1	The legend scale is fixed per measure, not auto-scaled to the current image, so that...

ID	§	Requirement
WF5.032	5.4.2	Side-by-side or swipe-divider comparison of the same plot on two dates, with both dates...
WF5.033	5.4.2	A compare with previous years mode showing the same calendar week across up to 5 previous years
WF5.034	5.4.3	Adding a new crop cycle to a plot that already has an open one prompts the user to close the...
WF5.035	5.4.3	Closing a crop cycle never deletes it
WF5.036	5.4.3	Crop cycle fields: crop, variety, sowing/planting date, expected harvest date, actual...
WF5.037	5.4.3	Yield here is what the farmer recorded, not a prediction
WF5.038	5.4.3	The crop picker is searchable, grouped by category (cereals, forage, vegetables, fruit...
WF5.039	5.4.4	A farm supports many plots with different crops, varieties and planting dates simultaneously
WF5.040	5.4.4	The hierarchy is farm → farm boundary → plot → tree
WF5.041	5.4.4	Plots sit directly under the farm boundary
WF5.042	5.4.5	A plot may record a primary crop and one or more secondary crops (for example date palm with...
WF5.043	5.4.5	Measure values and health status on an intercropped plot are attributed per crop
WF5.044	5.4.5	Recommendations are generated per crop within the plot
WF5.045	5.4.5	Where separation cannot be computed for a given date (e.g. insufficient resolution, cloud,...
WF5.046	5.4.6	Editable: farm name, farm type, address/region, soil type, default language for reports,...
WF5.047	5.4.6	Farm settings offers Run a survey on this farm for a farm created by hand, and Add a plot by...
WF5.048	5.4.6	Contains Transfer ownership (Owner only, with SMS confirmation) and Delete farm (Owner only,...
WF5.049	5.4.7	Add farm runs the same sequence as A9: draw the plots, or send a boundary for survey
WF5.050	5.4.7	Adding a sixth farm shows the enterprise screen first (WF4.104), with the option to continue...
WF5.051	5.4.7	Adding an eleventh farm is refused with a plain explanation and the enterprise contact route...
WF5.052	5.4.7	A farm added to an account of the other type follows WF4.109: visible, boundary kept,...
WF5.053	5.5	The screen leads with the status distribution; the list below is filtered to what needs...
WF5.054	5.5	Filters are status (healthy, declining, missing) and plot
WF5.055	5.5	The tree list creates no tasks
WF5.056	5.5	Tree detail (B10) shows: tree ID, tree type, its plot and row, a locator view (§5.7.1), the...
WF5.057	5.5	Planting year is not carried on the tree record
WF5.058	5.5	Tree detail has no create-task action (§5.8.1)
WF5.059	5.5	Missing and dead trees are shown as a distinct state with their own count, not folded into...
WF5.060	5.5	Where a tree-level measure is not in the client's plan, it is listed and locked rather than...
WF5.061	5.5	Analysis is per tree and per plot
WF5.062	5.6	Workforce is a named feature with its own screens
WF5.063	5.6	A worker record carries exactly four fields: name, mobile number, language and notification...
WF5.064	5.6	Creating a worker record does not create an app account and does not send an invitation
WF5.065	5.6	Notification preference is per person, a yes/no toggle for SMS and a yes/no toggle for WhatsApp
WF5.066	5.6	Work sent to a worker record goes out as a plain message in that worker's language, carrying...
WF5.067	5.6	A worker record can be promoted to an app account later by sending an invitation to the same...
WF5.068	5.6	Where a worker does have the app, their chosen app language overrides the language on the...
WF5.069	5.6	Workers who receive only SMS or WhatsApp never see a dashboard
WF5.070	5.6	A worker record can be deactivated
WF5.071	5.7	The map opens on the user's farms, zoomed to fit all of them, or to the currently selected...
WF5.072	5.7	Plot polygons are filled with the currently selected measure and carry their status icon and...
WF5.073	5.7	Tapping a plot opens a bottom sheet (C3) with: plot name, crop, area, current measure value...
WF5.074	5.7	Layer picker (C2) offers, subject to plan:
WF5.075	5.7	Layer selections persist between sessions
WF5.076	5.7	Locked layers appear in the list with a lock icon and open the upgrade screen when tapped

ID	§	Requirement
WF5.077	5.7	The user's own GPS position is shown when location permission is granted, with a Locate me...
WF5.078	5.7	Date stepper on the map moves through available imagery dates for the visible plots, as WF5.023
WF5.079	5.7	Compare mode (C4) puts a draggable vertical divider across the map with a different date...
WF5.080	5.7	Map tiles for the user's own farms are cached for offline use (§11.2)
WF5.081	5.7	The search bar is visible at all times on the map
WF5.082	5.7	The map header carries the name of the measure currently displayed
WF5.083	5.7	A Find a tree control sits on the map itself, because that is where the farmer is standing...
WF5.084	5.7	The farm selector on the map offers All farms as well as each farm individually
WF5.085	5.7.1	Show me where opens the map centred on the target plot or tree, with a line and a distance...
WF5.086	5.7.1	Identification in the field is by GPS proximity and heading
WF5.087	5.7.1	Where GPS accuracy is worse than the tree spacing, the app says so plainly and falls back to...
WF5.088	5.7.1	Tags and QR codes on trees are deferred, not rejected
WF5.089	5.7.1	The locator view works offline from cached tiles (§11.2) and states the age of the imagery...
WF5.090	5.7.2	Editing an existing boundary uses the same interaction as A9: tap to add, drag to move, and...
WF5.091	5.7.2	The boundary editor also splits one plot into two, joins two plots into one, removes a plot...
WF5.092	5.7.2	A boundary change is a versioned event, not an overwrite
WF5.093	5.7.2	Boundary editing requires the farm.boundary.edit capability (Owner and Supervisor)
WF5.094	5.8	Recommendations are grouped by urgency into Today / This week / Later, and within each group...
WF5.095	5.8	Every recommendation card states, in this order and in plain language: what to do, how much,...
WF5.096	5.8	Every recommendation carries four actions, reachable without opening the detail screen:
WF5.097	5.8	There is no interval menu on the reminder
WF5.098	5.8	An ignored item disappears from the inbox and returns the next day if the condition still holds
WF5.099	5.8	Every advisory item arrives pre-packaged as a task: description, timing and a suggested...
WF5.100	5.8	Before confirming, the farmer sees what they are approving and why — the instruction that...
WF5.101	5.8	Tapping the body of a card opens the detail view: why it was suggested, what assumptions...
WF5.102	5.8	Filters: farm (including All farms), plot, type (irrigation, nutrition, protection,...
WF5.103	5.8	A recommendation is marked done when the user records an action against it (D7) or completes...
WF5.104	5.8	Recommendations are never silently deleted or replaced
WF5.105	5.8	Where the client's plan does not include advisory, the Advice tab is still present and shows...
WF5.106	5.8.1	Advisory creates tasks
WF5.107	5.8.1	The task screen (E1) keeps an ADD button at the bottom right for the manual case
WF5.108	5.8.1	Either an owner, a supervisor or a worker may add a manual task
WF5.109	5.8.1	A task may be completed by the farm owner or the assigned worker
WF5.110	5.8.1	Manual task creation is expected to be rare
WF5.111	5.8.2	Irrigation advice states how much water the plot or the tree needs today, not how much water...
WF5.112	5.8.2	The recommendation carries a volume and a time: "93 m³ at 6 p.m."
WF5.113	5.8.2	Water is shown in cubic metres only
WF5.114	5.8.2	Times are shown on a 12-hour clock with a.m. and p.m
WF5.115	5.8.2	Pumping time is calculated and displayed where the plot has a recorded system flow rate
WF5.116	5.8.2	The "Why" section lists every input used, with its value
WF5.117	5.8.2	Assumptions we used is editable
WF5.118	5.8.2	The disclaimer line is present on every advisory detail screen, in the user's language, and...
WF5.119	5.8.3	Same structure: headline dose, split across application windows, the reasoning, editable...
WF5.120	5.8.3	Nutrient recommendations are expressed by element — N, P, K, Ca, Mg and the rest — per unit area
WF5.121	5.8.4	Recommendations lead with the active ingredient and the rate, never a commercial product name

ID	§	Requirement
WF5.122	5.8.4	Product names are drawn from the registered-product register for the user's country — see §6.4
WF5.123	5.8.4	Every crop protection recommendation displays the pre-harvest interval for the user's...
WF5.124	5.8.4	A permanent, non-dismissible notice: Check the product label and your local regulations...
WF5.125	5.8.4	The screen includes symptom photographs and a short identification guide from MMC's disease...
WF5.126	5.8.4	Every crop protection recommendation is written to the advisory log when it is generated —...
WF5.127	5.8.5	Alert types: frost, heat stress, high wind (spraying), heavy rain, sandstorm/dust, high...
WF5.128	5.8.5	Weather alerts are delivered as push notifications at the criticality defined in §7.2, in...
WF5.129	5.8.6	Recording an action takes at most three taps from the advice card for the common case
WF5.130	5.8.6	The recorded action writes to the plot's activity history and the farm diary, and is...
WF5.131	5.8.6	Once actions have been recorded for a plot, the plot detail screen shows an...
WF5.132	5.8.6	This screen works offline
WF5.133	5.9	Task states: Open → In progress → Done, plus Cancelled
WF5.134	5.9	A task has: title, description, type, farm, plot or tree selection, assignee, due date and...
WF5.135	5.9	The ADD button sits at the bottom right of the task list and is the only manual...
WF5.136	5.9	On the task screen the assigned person is at the very top, the edit control at the bottom,...
WF5.137	5.9	Task types (used for icons, filtering and reporting): irrigation, fertiliser, spraying,...
WF5.138	5.9	A task may be assigned to one person, who may be an app user or a §5.6 worker record
WF5.139	5.9	A task may cover many plots or many trees
WF5.140	5.9	Overdue tasks are always shown first, always visually distinct
WF5.141	5.9	Scope: light task management, not a farm management system
WF5.142	5.9	Where the platform can determine a schedule — irrigation timing, spray windows, harvest...
WF5.143	5.9.1	The screen opens fully pre-filled when it comes from advisory, assignee included where the...
WF5.144	5.9.1	The line under the save button names the language the assignee will receive the task in,...
WF5.145	5.9.1	Creating a task from a recommendation pre-fills title, description (including what and how...
WF5.146	5.9.1	The assignee picker lists app users with access to the selected farm and worker records held...
WF5.147	5.9.1	On save, the assignee is notified through their configured channel (§7) in their language,...
WF5.148	5.9.2	The Worker view of a task is stripped to: what, how much, where, when, and one big button
WF5.149	5.9.2	Mark as done opens E4
WF5.150	5.9.2	Everything on the completion screen is optional except the confirm tap
WF5.151	5.9.2	Photographs are automatically stamped with GPS coordinates, timestamp, task ID, plot ID and...
WF5.152	5.9.2	The whole completion flow works offline, confirms locally and immediately, and syncs later (§11)
WF5.153	5.9.2	I cannot do this records a reason from a short picture-supported list and notifies the task...
WF5.154	5.9.3	Category and severity are chosen from labelled picture buttons, and only the photograph is...
WF5.155	5.9.3	Severity is expressed in plain words : a few, patchy, widespread
WF5.156	5.9.3	A standalone observation can be captured from the Tasks tab, the map, a plot or a tree,...
WF5.157	5.9.3	Observations appear on the map as pins, on the plot's activity history, and in the farm diary
WF5.158	5.9.3	Where the plan includes photo-based disease detection (E7), photographing a leaf returns an...
WF5.159	5.9.3	Works fully offline
WF5.160	5.10	Menu contents are filtered by role
WF5.161	5.10	The version and build number are always visible on this screen
WF5.162	5.10.1	Reports are generated server-side as PDF. Excel export is offered for the tabular reports
WF5.163	5.10.1	Every report is produced in the language the user requests, from the same string catalogue...
WF5.164	5.10.1	Reports are shareable through the OS share sheet (WhatsApp, email, file save)
WF5.165	5.10.1	The Irrigation: advised vs applied report is not in the existing report set and must be built
WF5.166	5.10.1	Reports outside the client's plan appear locked

ID	§	Requirement
WF5.167	5.10.1	Two summary reports sit alongside the weekly farm report: a health summary across plots and...
WF5.168	5.10.2	Pending invitations are listed with their expiry and a one-tap cancel
WF5.169	5.10.2	Each member row shows their role and their app language, so the assigner knows which...
WF5.170	5.10.2	Invite (F3) is by mobile number (primary), email, link, 6-character code or QR — see §4.1
WF5.171	5.10.2	Invitations may be scoped to specific farms, not necessarily the whole account
WF5.172	5.10.2	QR-code invitations produce single-use, short-expiry, role-scoped tokens
WF5.173	5.10.2	Removing a member takes effect immediately, invalidates their active sessions for that farm,...
WF5.174	5.10.3	An account holds one subscription covering every farm on it, up to the 10-farm cap
WF5.175	5.10.3	Trial status shows days remaining, prominently, from day one
WF5.176	5.10.3	Where the subscription was bought in the app, purchase and renewal run through Apple In-App...
WF5.177	5.10.3	Entitlement is always determined by the server, never by the local receipt alone, and never...
WF5.178	5.10.3	Where the subscription was bought on the web, this screen states that billing is managed...
WF5.179	5.10.3	The Contact Wafra route for invoicing, annual contracts and team seats is informational only...
WF5.180	5.10.4	Language, units and region (F8): app language; land unit (dunum, hectare, acre); distance...
WF5.181	5.10.4	The unit choice made at A7 (§4.8) is the starting value here and can be changed at any time
WF5.182	5.10.4	The setting covers the measurement system generally, not area alone
WF5.183	5.10.4	Notifications (F9): per-channel and per-category toggles, quiet hours, and a channel...
WF5.184	5.10.4	Data and storage (F10): current cache size, "download imagery on Wi-Fi only" toggle, "clear...
WF5.185	5.10.4	A shared device toggle, which shortens the session to 12 hours and requires...
WF5.186	5.10.4	Settings includes Delete my account, which explains what will be deleted and what will be...
WF5.187	5.10.5	Owner-only
WF5.188	5.10.5	The activity log is append-only
WF5.189	5.10.6	Help contains a searchable user guide, in all five languages, with short illustrated articles
WF5.190	5.10.6	Contact Wafra presents exactly two large buttons plus a support ticket option:
WF5.191	5.10.6	WhatsApp opens a chat with +966 54 810 0443, pre-filled with a short message containing the...
WF5.192	5.10.6	Email opens the device mail client to info@wafra.green.com, with the same diagnostic details...
WF5.193	5.10.6	Both destinations are configuration values fetched from the backend, not compiled into the...
WF5.194	5.10.6	The support ticket route uses the existing ticketing capability where the plan includes it
WF6.001	6.3	Every recommendation is written to an advisory log at the moment it is generated, before it...
WF6.002	6.3	The advisory log is write-once
WF6.003	6.3	Retention is 7 years
WF6.004	6.3	A superseded recommendation is marked superseded and linked to the one that replaced it
WF6.005	6.3	The log is exportable, on request, for a named client and date range, in a documented...
WF6.006	6.3	Acknowledgement and action are recorded against the log entry: who saw the advice, when, and...
WF6.007	6.4	Wafra owns the product register — plant protection products, nutrition products and their...
WF6.008	6.4	Wafra maintains the register over time
WF6.009	6.4	Each change to the register content is handled as a change order to MMC, with its own record
WF6.010	6.4	A register of plant protection products, held per country, containing at minimum: active...
WF6.011	6.4	A product is only shown to a user whose farm is in a country where that product is registered
WF6.012	6.4	Recommendations are expressed by element or active ingredient — N, P, K, calcium and the...
WF6.013	6.4	The pre-harvest interval is taken from the register for the user's country, and the earliest...
WF6.014	6.4	Register entries are versioned with an effective-from date
WF6.015	6.4	The register is updated server-side and takes effect without an app release
WF6.016	6.4	Every change to the register is recorded with who changed it, when, and what changed
WF6.017	6.4	Where an entry has not been verified within 12 months, product names from that entry are...

ID	§	Requirement
WF6.018	6.5	Every recommendation carries a rule_version
WF6.019	6.5	Every input used is stored with the recommendation, with its value, unit and source
WF6.020	6.5	Assumptions a user can correct — irrigation efficiency, soil type, system flow rate — are...
WF6.021	6.6	Where the client's plan does not include a given advisory capability, no recommendation of...
WF6.022	6.6	Where canopy separation cannot be computed for an intercropped plot on a given date,...
WF6.023	6.6	Where a mandatory input is missing, no recommendation is generated with a substituted value
WF6.024	6.6	Crop protection recommendations above a defined risk threshold — rate at the top of the...
WF6.025	6.7	Every advisory detail screen carries a non-dismissible line stating that the recommendation...
WF6.026	6.7	Every crop protection recommendation carries a non-dismissible instruction to check the...
WF6.027	6.7	The Terms of Use state that recommendations are advisory, that the farmer remains...
WF6.028	6.7	Advice is never presented as certain where it is not
WF7.001	7.1	Three channels: push notification, WhatsApp, and SMS
WF7.002	7.1	Every notification is sent in the recipient's saved language, not the sender's, not the...
WF7.003	7.1	WhatsApp delivery uses the WhatsApp Business API with pre-approved message templates
WF7.004	7.1	SMS is the fallback where WhatsApp delivery fails or the recipient has not opted in
WF7.005	7.1	For a worker record the channel is not a fallback chain but an explicit per-person choice:...
WF7.006	7.2	Categories marked "no" above are safety- or contract-critical and may not be switched off...
WF7.007	7.2	Quiet hours are user-configurable, default 9 p.m. to 5 a.m. local, and apply to every...
WF7.008	7.2	Every notification carries a deep link that opens the exact object it concerns — task,...
WF7.009	7.2	Notification volume is capped at 6 per user per day, excluding task assignments and severe...
WF8.001	8.1	Permissions are implemented as a capability matrix
WF8.002	8.1	Capabilities are the unit of enforcement, not role names
WF8.003	8.2	A Supervisor may invite Workers but not other Supervisors or Owners
WF8.004	8.2	The structure to model is owner → supervisor → a small number of workers
WF8.005	8.2	worker.manage covers creating, editing and deactivating worker records (§5.6) and is...
WF8.006	8.2	A Worker may create a manual task, but may not assign one to anyone else (§5.8.1)
WF8.007	8.2	Access is granted per farm, not per account
WF8.008	8.2	A farm may have more than one Owner
WF8.009	8.3	Every capability is enforced server-side
WF8.010	8.3	Session tokens carry the user identity only
WF8.011	8.4	Invitation tokens are single use, expire after 7 days, are bound to the role and farm scope...
WF8.012	8.4	An invitation link that has been used, expired or revoked shows a clear message and a...
WF9.001	9.1	Two levels, basic and pro, in three services
WF9.002	9.1	Plan names in the interface are exactly Basic and Pro
WF9.003	9.1	Every feature is either "basic and pro" or "pro only"
WF9.004	9.1	Wafra's names and MMC's names for the same tier are different and must be reconciled in...
WF9.005	9.1	The mapping above is server-side configuration
WF9.006	9.1	Nothing that exists only in MMC's Advanced tree tier is sold
WF9.007	9.1	The exclusion is tier-specific, not feature-name-specific
WF9.008	9.1	The app does not estimate yield
WF9.009	9.1	The subscription sits at farm level and resolves to one account-level entitlement
WF9.010	9.1.1	Price is computed from what the survey detected, never from a static list: crop capability...
WF9.011	9.1.1	Rates, currencies and any volume banding are server configuration, fetched at launch and on...
WF9.012	9.1.1	Volume banding on the supplier side is aggregated across all Wafra clients, not per end...
WF9.013	9.1.1	A price is never displayed for a farm whose survey has not completed and been confirmed...

ID	§	Requirement
WF9.014	9.1.1	Crop area is priced per acre or per hectare and tree capability per tree
WF9.015	9.1.2	An account holds one plan at a time, whatever mix of farms it contains
WF9.016	9.1.2	The service offered is derived from the account's farms, not chosen by the user: crop farms...
WF9.017	9.1.2	Adding a farm of the other type to a single-service account puts that farm in a pending...
WF9.018	9.1.2	Archiving every farm of one type offers a downgrade from Combined to the remaining single...
WF9.019	9.1.2	Trees are counted for entitlement at the account level across all tree farms, and crop area...
WF9.020	9.1.3	Path A — in-app purchase
WF9.021	9.1.3	Path B — web purchase with a server-side entitlement
WF9.022	9.1.3	Path B depends on the account carrying an email address, which is why email is mandatory on...
WF9.023	9.1.3	The app must not advertise, describe, link to or hint at the web purchase path from inside...
WF9.024	9.1.3	The registration flow leads to Path A. A web purchase produces no purchase screen in the app...
WF9.025	9.1.3	The app reads entitlement state at launch and on resume, caches the last known state for...
WF9.026	9.1.3	Offline, the app honours the last entitlement state it holds and shows its age
WF9.027	9.1.3	Path C — government and bulk
WF9.028	9.1.3	A seat provisioned under Path C is entitled by the same server-side mechanism as Path B. The...
WF9.029	9.2	Every new account receives a 30-day free trial on the plan they select at A13, with the full...
WF9.030	9.2	Trial state is held server-side
WF9.031	9.2	Trial expiry warnings at 7 days, 3 days and 1 day remaining, by push and email
WF9.032	9.2	On trial expiry without a subscription, the account enters read-only state: the user can...
WF9.033	9.5	A feature outside the user's plan is visible and locked, never hidden, unless it would be...
WF9.034	9.5	A locked feature shows a lock icon in place of its value or content
WF9.035	9.5	The upgrade sheet names the plan required and describes the benefit in one sentence of plain...
WF9.036	9.5	Entitlement checks happen server-side on every request
WF10.001	10.1	Five languages ship at launch: Arabic, English, Hindi, Bengali, Pashto
WF10.002	10.1	In the language picker each language appears in its own script only — no English gloss, no...
WF10.003	10.1	Arabic and Pashto are both right-to-left
WF10.004	10.1	RTL treatment includes: layout mirroring, navigation and back-arrow direction, list...
WF10.005	10.1	Numbers use Western Arabic numerals (0–9) in all languages including Arabic, unless the user...
WF10.006	10.1	Fonts covering Devanagari (Hindi), Bengali and Perso-Arabic (Pashto) are bundled with the...
WF10.007	10.1	Test on a physical Android 8 device in each script before sign-off
WF10.008	10.1	Known rendering faults to fix and to test for explicitly: an English label repeated beneath...
WF10.009	10.1	Language switching takes effect immediately across the entire interface, without restarting...
WF10.010	10.1	Layouts accommodate a 40% expansion in string length without breaking
WF10.011	10.2	All strings live in an i18n platform with stable keys
WF10.012	10.2	An agronomic glossary is maintained alongside the string catalogue, fixing the agreed...
WF10.013	10.2	A named reviewer per language is recorded and signs off the catalogue before release
WF10.014	10.2	Server-generated content — recommendation text, alert bodies, report sections — comes from...
WF10.015	10.3	Every notification, task instruction, report and recommendation is rendered in the...
WF10.016	10.3	Where a translation is missing for a key, fall back to English and log the missing key...
WF10.017	10.4	Temperature in Celsius
WF10.018	10.4	Times are shown on a 12-hour clock with a.m. and p.m
WF10.019	10.4	The land unit is asked at sign-up and defaults from the country detected on the phone —...
WF10.020	10.4	Dates display in both Gregorian and Hijri when the app language is Arabic, Gregorian...
WF10.021	10.4	Prices display in the user's local currency — the GCC currencies and the Jordanian dinar —...
WF10.022	10.4	Time zones: all timestamps stored in UTC, displayed in the farm's local time zone, which is...

ID	§	Requirement
WF11.001	11.1	The app is read-anywhere, write-in-the-field
WF11.002	11.2	Total cache is capped at 500 MB by default, user-adjustable between 100 MB and 2 GB in Settings
WF11.003	11.2	Queued user data (photos, completions, observations) is never evicted and does not count...
WF11.004	11.3	Actions that cannot be performed offline show a clear, calm message naming what is needed...
WF11.005	11.4	Every offline-created record is written locally with a client-generated UUID and an...
WF11.006	11.4	Sync is automatic on regaining connectivity, on app foreground, and on a manual Sync now
WF11.007	11.4	Conflict rules, per object type:
WF11.008	11.4	A sync failure is surfaced to the user in the connectivity indicator and in Settings → Data...
WF11.009	11.4	Photos queued for upload survive the app being force-closed, the device being restarted and...
WF11.010	11.4	Upload retries use exponential backoff, resume partial uploads, and prefer Wi-Fi where the...
WF11.011	11.4	The offline store, including cached imagery and queued photos, is encrypted at rest on the...
WF11.012	11.5	A connectivity indicator is present on every screen, with three visual states: online,...
WF11.013	11.5	Cached content older than 24 hours carries a visible age label